

Unit 11 Overview: Operations With Rational Numbers

Unit Narrative



In Unit 11, students begin working on their fluency with operations with rational numbers in mathematical contexts. Students then extend this to problem solving in real-world contexts with all rational numbers. In Unit 6, students were focused on fluency with operations with positive rational numbers and with integers, as well as problem solving in real-world contexts with these numbers. This extends that fluency to include all rational numbers. Students will also practice performing operations with rational numbers through the process of evaluating expressions using the order of operations.

In Section 1 of this unit, students will be exposed to operations with rational numbers written as numeric expressions. Many of the problems that students will solve involve performing operations given just two rational numbers. Section 1 also engages students in finding the absolute value and applying the order of operations. In Section 2 of the unit, students will be solving problems that involve more than one operation. Section 2 also exposes students to operations with rational numbers in a wide variety of real-world scenarios, building on the skills learned in Unit 2 when solving real-world problems involving positive rational numbers.

Section 3 has a primary focus on evaluating expressions using the order of operations. In this section, students will apply what they have learned about operations with rational numbers to evaluate expressions that include exponents and absolute value. They will revisit absolute value notation from Unit 3 and nested grouping symbols.

Mathematical Thinking and Reasoning (MTR) Standards 3 (MA.K12.MTR.3.1), 4 (MA.K12.MTR.4.1) and 7 (MA.K12.MTR.7.1) drive instruction in this unit. In lessons 5-7, MA.K12.MTR.4.1 engages students in communicating mathematical ideas and analyzing the mathematical thinking of others while MA.K12.MTR.7.1 incorporates real-world connections for students to establish more meaningful engagement in problem-solving. Additionally, there is a focus on developing fluency with operations as the Grade 7 fluency benchmarks are included in this unit. Students are expected to be fluent in performing operations with integers and rational numbers by the end of this course, which means they have developed efficient and accurate procedures.

Unit At-A-Glance

Section 1: Operations and Rational Numbers	Benchmarks Addressed	Lesson 1	Adding Rational Numbers
	MA.7.NSO.2.2	Lesson 2	Subtracting Rational Numbers
		Lesson 3	Multiplying Rational Numbers
		Lesson 4	Dividing Rational Numbers
Section 2: Operations and Rational Numbers in the Real World	Benchmarks Addressed	Lesson 5	Solving Real World Problems With Rational Numbers – Part 1
	MA.7.NSO.2.3	Lesson 6	Solving Real World Problems With Rational Numbers – Part 2
		Lesson 7	Solving Real World Problems With Rational Numbers – Part 3
Section 3: Evaluating Numerical Expressions	Benchmarks Addressed	Lesson 8	Evaluating Numeric Expressions – Part 1
	MA.7.NSO.2.1 MA.7.NSO.2.2	Lesson 9	Evaluating Numeric Expressions – Part 2



Differentiation

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Engagement <i>The “WHY” of learning</i>	- Lessons 5-7 incorporate real-world contexts for students to engage with mathematics in meaningful ways. Cultural intelligence and sociocultural capital varies by community and context; therefore, some real-world contexts may not fully reflect the “real” worlds of your students and classroom. Consider engaging students by adjusting or altering contexts and scenarios to reflect their local contexts and communities. Alternatively, allow students to alter the problems themselves, substituting people, places, or purposes, while retaining the numbers to be able to still discuss common misconceptions. - Alternatively, real-world problems also afford opportunities to discuss real-world solutions. For example, one scenario from Lesson 7 centers on an energy company and compares traditional to solar power. Consider engaging students with contemporary conversation starters for meaningful discussion. For example, should all homes be required to switch to solar energy? Why or why not?
Representation <i>The “WHAT” of learning</i>	Performing operations with rational numbers involves a plethora of calculations, and not all students calculate the same way. Synesthesia, for example, is a neurological condition in which information designed for one sense (like calculations) stimulates multiple senses. A student with synesthesia might struggle with multiplication recall and instead sees multiples as calendar months or colors. Students most often exhibit or communicate this intelligence as spatial-sequencing. Make space for students to describe how they best represent operations with rational numbers, and you may be surprised with what you learn about your students.
Action & Expression <i>The “HOW” of learning</i>	- Provide manipulatives, such as two-color counters, to assist students with demonstrating calculations. - Consider color coding the different forms of rational numbers. Post rules and reminders about performing operations with a specific type of rational number in a particular color. Encourage students to complete all calculations in that particular color. This will assist students with connecting the processes that go along with the type of rational number. - Offer project-based learning experiences for students to replicate the problem-solving of Lessons 5-7 in meaningful contexts. Students often demonstrate understanding of operations either at home or in their community, unknowingly performing multiple operations. Projects provide intentional space for students to connect the math they are doing to metacognitive approaches to mathematics in the classroom.

ELL Information

Mark-Up Text

In Section 2 students will be solving real-world problems using operations with rational numbers. This section is text and reading heavy. When students translate words into numbers, expressions, or equations, encourage them to use a highlighter to highlight phrases and words to call attention to keywords. Jose's video will be helpful for ELL students to watch for this topic because he will model these strategies. If ELL students respond well, consider recommending that Jose be their go-to Study Expert!

Because these sections are text-heavy, "Mark-Up Text" is a good activity to help ELL students begin feeling comfortable translating words into mathematical expressions.

- Teacher will use a word problem where students will be able to identify which parts would need to be represented in numbers, and how to write those numbers in an expression or equation.
- Teacher will provide students with the following symbols.
 - ✓ - background knowledge, they know what to do
 - ? - confusing information; is this needed to solve the problem?
 - ! - marks where a variable is needed to represent a number
 - + - extra info in word problem; not needed
- Using a small piece of text, teacher will model for students how you would Mark Up before releasing them to do the same.
- Students will read the text slowly and carefully, focusing on one sentence at a time and marking text based on the key. Then they will turn to a partner to discuss each sentence and compare where they made marks.
- For the first few examples, have groups share their decisions out to the class. Then, guide students through turning the marked-up text into an expression. ELL students may find this a helpful intermediary step, so encourage them to use this strategy even as you release them to work independently.

All glossary terms are available in multiple languages, including English, Spanish, Haitian Creole, and American Sign Language, as support for ELL students.

Additional Differentiated Support

Scaffolding Support	On-Ramp
Study Expert videos can function as an additional co-teacher or additional scaffolding for specific operations or contexts. Consider utilizing resources from Grade 6 Accelerated Units 1, 2, and 6 to support students who may need additional support or remediation with performing operations. This support would be similar to using On-Ramp videos but tailored to activities within the curriculum. Consider an example using stations during a lesson. • Station 1: Students work on Your Turn! or Try It practice problems independently. • Station 2: Teacher facilitates a small group in which the content is determined by the group. • Station 3: Students can choose their level of support, either watching a Study Expert from Grade 6 Unit 2 or engaging in the unit extension activity.	On-Ramp to 6th grade provides video and practice tools for students to scaffold the following skills: • Evaluate Expressions • Addition Fluency • Subtraction Fluency • Adding Decimals • Subtracting Decimals • Multiplying Decimals • Dividing Decimals • Add Mixed Numbers with Like Denominators • Subtract Mixed Numbers with Like Denominators • Add Fractions with Unlike Denominators • Subtract Fractions with Unlike Denominators • Number Lines: Whole Number Sums and Differences • Number Lines: Mixed Numbers

- Number Lines: Fractions
- Fractions in Real-World Addition and Subtraction Problems
- Fractions in Real-World Multiplication Problems
- Adding Two-Digit and One-Digit Numbers



Section Overview

Section 1: Operations and Rational Numbers (Lessons 1-4)

Benchmarks Addressed	Learning Targets	
MA.7.NSO.2.2 Add, subtract, multiply and divide rational numbers with procedural fluency.	Lesson 1	• I can fluently add rational numbers.
	Lesson 2	• I can fluently subtract rational numbers.
	Lesson 3	• I can fluently multiply rational numbers.
	Lesson 4	• I can fluently divide rational numbers.

Section 2: Operations and Rational Numbers in the Real World (Lessons 5-7)

Benchmarks Addressed	Learning Targets	
MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers. • <i>Clarification 1: Instruction includes using one or more operations to solve problems.</i>	Lesson 5	• I can solve problems with one or two operations by adding, subtracting, multiplying, and/or dividing rational numbers.
	Lesson 6	• I can solve problems with three operations involving any of the four operations with rational numbers.
	Lesson 7	• I can solve problems with at least three operations involving any of the four operations with rational numbers.

Section 3: Evaluating Numerical Expressions (Lessons 8-9)

Benchmarks Addressed	Learning Targets	
MA.7.NSO.2.1 Solve mathematical problems using multi-step order of operations with rational numbers including grouping symbols, whole-number exponents and absolute value. • <i>Clarification 1: Multi-step expressions are limited to 6 or fewer steps.</i>	Lesson 8	• I can evaluate numeric expressions with up to four operations.
	Lesson 9	• I can evaluate numeric expressions with up to six operations.
MA.7.NSO.2.2 Add, subtract, multiply and divide rational numbers with procedural fluency.		



Vertical Alignment

Grade 6 Accelerated Unit 11
Operations with Rational Numbers



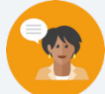
Grade 7 Accelerated Unit 4
Solving Problems with Rational Numbers

Prior Course(s)	Course Level	Future Course(s)
	MA.7.NSO.2.1 Solve mathematical problems using multi-step order of operations with rational numbers including grouping symbols, whole-number exponents, and absolute value. MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency. MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers.	MA.8.NSO.1.5 Add, subtract, multiply and divide numbers expressed in scientific notation with procedural fluency. MA.8.NSO.1.6 Solve real-world problems involving operations with numbers expressed in scientific notation MA.8.NSO.1.7 Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers including exponents and radicals.

11.1 Adding Rational Numbers

Lesson Introduction

Lesson Introduction				
Benchmark	MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.		Learning Target	I can fluently add rational numbers.
Benchmark Coherence	Building On		Working Toward	
	MA.6.NSO.4.1 Apply and extend previous understandings of operations with whole numbers to add and subtract integers with procedural fluency.		MA.8.NSO.1.7 Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers including exponents and radicals.	
Essential Questions	- How is the process of adding integers apply to adding rational numbers? - How is the sign of the sum determined when adding rational numbers?			
Lesson Narrative	This lesson opens with a video about a landscape architect where collaboration is an apparent part her job. Next, the lesson presents adding decimals, fractions, and integers separately, as well as investigating the process of each. The lesson continues with adding mixed numbers and describing the process. Students practice adding rational numbers, and the lesson wraps up with students selecting their own fraction or mixed number and finding the sum.			
Component Summary	11.1.1 Warm-Up (~5 min.)	Career Profile Video (<i>SE p. 165</i>)	11.1.4 Refresher (15 min.)	Add and subtract rational numbers and complete statements (<i>SE p. 167</i>)
	11.1.2 Guided Instruction (10 min.)	Analysis of student work of adding decimals and fractions (<i>SE p. 166</i>)	11.1.5 Guided Practice (10 min.)	Complete addition expressions (<i>SE p. 169</i>)
	11.1.3 Try It (5 min.)	Pair equivalent rational expressions (<i>SE p. 167</i>)	11.1.6 Your Turn! (5-10 min.)	Find the sum of rational number expressions (<i>SE p. 170</i>)
	11.1.7 Wrap-Up (~5 min.)	Add rational numbers (<i>SE p. 170</i>)		
Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - N/A <u>Additional Terms:</u> - Absolute value		- Career Profile Video - Calculators (only for question 7 in 11.1.4)	11.1.1 Warm-Up contains a video to accompany the questions.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may not convert the numerators after changing the denominators to the common denominator of fractions. - Students may not include the whole number when adding mixed numbers and fractions.	- Students may write the common denominator and the new numerators adjacent to the original fraction. Consider advising the students to rewrite the equivalent fractions underneath the original fractions to assist with organization and clarity. - Determine if your students need the support of all or part of the refresher. - If time is an issue, then consider assigning 11.1.6 Your Turn! for homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Compare and Connect 11.1.5 Guided Practice presents students with adding two fractions and then adding two mixed numbers. Consider using a compare and contrast routine where the students compare and contrast the differences and similarities between the processes for adding fractions and adding mixed numbers.	MA.K12.MTR.3.1: Mathematical Fluency Mathematical Fluency covers flexibility and precision while carrying out calculations. By providing students with multiple opportunities to execute procedures and perform calculations, this lesson allows students to maintain flexibility and accurately add a variety of rational numbers.
 Differentiate Instruction	11.1.3 Try It and 11.1.6 Your Turn! both provide practice opportunities for students to engage with a variety of tools and resources that provide differentiated support. Students can engage with On-Ramp videos on the following topics for additional support: - Adding Decimals - Add Fractions with Unlike Denominators	
Lesson Notes:		

11.1.1 Warm-Up: Landscape Architect

Component Overview: Play the Career Profile video on the Landscape Architect for the entire class. After students watch the video (3:18 minutes), prompt them to reflect on the profession by answering the two questions independently.



11.1.1 Warm-Up: Landscape Architect

Janice Nicol, a landscape architect, works to design natural systems and outdoor spaces, like trails and parks, for campuses and recreational facilities. Her job is both creative and technical. Creativity is expressed through drawing sketches both by hand and by using computer design programs and imaging. The technical side of her job involves knowing things such as land grading, water calculations, and natural systems.

Janice mentions that she regularly collaborates and works together with others while doing her job.

1. Why do you think it is important to be able to collaborate when doing a job?
Answers will vary. It is important to collaborate because different people have different ways of thinking that can help get the job done quicker.

2. How can you build your collaboration skills?
Answers will vary. You can build collaboration skills by asking others their opinions or asking what their thought process is on a specific thing.



In order to assist students with crafting complete responses, post a thinking map, such as a circle map, of key characteristics of a collaborative team member. Some of these characteristics can include being a good listener or a creative contributor toward team goals.

11.1.2 Guided Instruction: Adding Positive Rational Numbers

Component Overview: Students complete an error analysis of student work to determine the mistake in adding decimals and correct it. Students then add fractions with different denominators and complete statements to explain how to add fractions with unlike denominators.



11.1.2 Guided Instruction: Adding Positive Rational Numbers

1. Adrian added 5.25 and 6.4. His work is shown.

Adrian's Work
$\begin{array}{r} 5.25 \\ + 6.4 \\ \hline 5.89 \end{array}$



Consider providing students with a place value chart to help align the decimals correctly to find the sum. This will provide a helpful visual for students to determine the error in Adrian's work.

- a. Write an explanation to Adrian detailing any mistakes he made and how he can correct his work.
Adrian, you did not line up your decimals to ensure the place values matched; this is necessary in order to add correctly.

- b. What is the sum of 5.25 and 6.4?
 11.65



When finding the sum of decimal numbers, add values with the same place value.

11.1.2 Guided Instruction: Adding Positive Rational Numbers (continued)

2. Johannes added $\frac{1}{8}$ and $\frac{3}{4}$. Part of her work is shown.

a. Complete Johannes' work by filling in the boxes.

Johannes' Answer
$\frac{1}{8} + \frac{3}{4} =$
$\frac{1}{8} + \frac{3}{4} \left(\frac{2}{2} \right) =$
$\frac{1}{8} + \frac{6}{8} = \frac{7}{8}$

b. Complete the statements by choosing a word or number from the bank. Not all words and numbers may be used, and some may be used more than once.

Word Bank		
2	4	8
numerators	common denominator	
one half	fourths	seven eighths

To find the sum of $\frac{1}{8}$ and $\frac{3}{4}$, Johannes first rewrote each fraction using a common denominator.

The common denominator of 8 and 4 that

Johannes used was 8. Then, she added the two

numerators together to find the total

number of seven eighths.



Prior to completing question 2b, encourage students to discuss the vocabulary in the Word Bank. Students may recall that the numerator is the number above the fraction bar in the fraction. Students might also point out that common denominators are the numbers below the fraction bar that are the same when adding two or more fractions.

11.1.3 Try It: Adding Positive Rational Numbers

Component Overview: Students will solve fraction expressions and will pair the expressions that result in the same sum.



11.1.3 Try It: Adding Positive Rational Numbers

1. Match pairs of expressions that have the same sum.
Write the equivalent expressions on the lines provided.

$\frac{3}{4} + \frac{4}{8}$	$\frac{2}{3} + \frac{3}{5}$	$\frac{2}{3} + \frac{7}{18}$
$\frac{5}{9} + \frac{1}{2}$	$\frac{15}{15} + \frac{8}{30}$	$\frac{12}{16} + \frac{1}{2}$

$$\underline{\frac{3}{4} + \frac{4}{8}} = \underline{\frac{12}{16} + \frac{1}{2}}$$

$$\underline{\frac{2}{3} + \frac{3}{5}} = \underline{\frac{15}{15} + \frac{8}{30}}$$

$$\underline{\frac{2}{3} + \frac{7}{18}} = \underline{\frac{5}{9} + \frac{1}{2}}$$



As you circulate listen for the strategies students use to determine their answers. Are they finding common denominators for each sum or are they simplifying fractions? Are they matching based on LCM's of the denominator? Invite students with different strategies to share them with their peers.



To provide students with a challenge, consider inquiring about one common denominator for all twelve fractions. This will give students the opportunity to find the least common denominator for a wide range of numbers.

11.1.4 Refresher: Operations with Integers

Component Overview: Students will work with a partner to find the sum of addition expressions and will answer questions regarding the process. Students will also verify solutions with a calculator and will complete statements about the process.



11.1.4 Refresher: Operations with Integers

1. Work with your partner to complete the table. The first addition expression has been done for you.

Expression	Absolute Value of First Addend	Absolute Value of Second Addend	Addend With Greater Absolute Value	Sign of Sum	Sum
$(-10) + 6$	10	6	-10	-	-4
$10 + (-6)$	10	6	10	+	4
$(-10) + (-6)$	10	6	-10	-	-16
$10 + 6$	10	6	10	+	16

2. How did the size of the absolute value for each addend determine the sign of the sum?
The sign of the addend with the greater absolute value is the sign that the sum will have.
3. Explain how finding the sum for the first two expressions was different from finding the sum for the last two expressions.
The first two expressions have addends with different signs; the answer is the difference between the absolute values of the addends and takes the sign of the addend with the larger absolute value. The last two expressions have addends with the same sign, and the answer is the sum of the absolute values of the addends and has the same sign as the addends.
4. How were the sums for $10 + (-6)$ and $(-10) + 6$ similar?
The absolute values of the sums are the same.



If students still struggle with operations with integers, then consider providing students with integer chips to assist them with completing the table for question 1. This manipulative will assist students with observing the sign of each integer.



Consider making connections to student responses to question 3 when completing statements in question 6. Students will likely provide statements similar to what is structured in question 6.

11.1.4 Refresher: Operations with Integers (continued)

5. How were the sums for $(-10) + (-6)$ and $10 + 6$ different?

The signs of the sums are different.

6. Complete the statements using the words *same* or *different* to describe the rules for finding the sums of signed numbers.

When two addends have the same sign, the sum will have the sign of the addends.

When two addends have different signs, the sum will have the sign of the addend with the greater absolute value.

7. Apply the rules for integers to these expressions with rational numbers. Find the sum of each expression. Then, verify your answer using a calculator.

Addition Expression	Sum When Using the Rule	Sum With a Calculator	Does the Rule Work?
$(-0.92) + 50.6$	49.68	49.68	Yes
$4.95 + 3.05$	8	8	Yes
$\left(-\frac{4}{5}\right) + \left(\frac{22}{25}\right)$	$\frac{2}{25}$	$\frac{2}{25}$	Yes
$1.25 + (-6.45)$	-5.20	-5.20	Yes
$\left(-\frac{2}{3}\right) + \left(-\frac{9}{20}\right)$	$-\frac{67}{60}$	$-\frac{67}{60}$	Yes
$(-17.01) + (-2.99)$	-20	-20	Yes

8. Complete the sentence.

The sign rules for adding integers are

- ☒ the same
☐ not the same

as the rules for adding rational numbers.



Prompt students to revisit their responses to question 3 when answering question 6.



Students are extending the rules they learned for operations with integers to the operations with rational numbers. Help students connect that the rules they already know are applied to rational numbers. Consider assigning each addition expression in the table to different sets of students.

Consider whether to verify the sums using a calculator as a whole group or whether to provide a calculator to certain students to verify. Calculators should not be used on any other part of the lesson.

Encourage the students to add on an additional column to the table to show their work. The space to provide work on the table will emphasize solving the questions using the rule first, instead of just using the calculator to find the sum.

11.1.5 Guided Practice: Applying Signed Number Rules to Rational Numbers

Component Overview: Students fill in boxes to make incomplete equations true. Students then collaborate with a partner to describe how to determine the missing value in equations.



11.1.5 Guided Practice: Applying Signed Number Rules to Rational Numbers

- The blanks in the addition equation represent missing values. Fill in the blanks for the addition equation below to make a true statement.

Answers will vary.

$$-12\frac{3}{4} + (-5\frac{\boxed{3}}{6}) = -18\frac{\boxed{3}}{12}$$

- Find the sum for each addition expression.

Addition Expression	Sum
$\frac{4}{5} + (-\frac{3}{7})$	$\frac{13}{35}$
$-9\frac{1}{2} + (-14\frac{1}{8})$	$-23\frac{5}{8}$



Prior to completing question 2, encourage students to compare and contrast the processes for adding and subtracting fractions versus the processes for adding and subtracting mixed numbers.

- The blanks in the addition equation represent digits in the values.

- Fill in the blanks for the addition equation below to create a true equation.

$$-16.537 + 6.1\boxed{7}8 = -10.\boxed{3}59$$

- Describe the process you used to determine the values for each of the blanks above.

I lined up the decimals to subtract. I figured out what the missing numbers were by finding numbers that would make the subtraction statement true.

- Discuss your process with your partner. Write a summary of their process. Have your partner initial to show that your summary accurately expressed their process.

Your Partner's Process

Answers will vary. I added 16.537 and $-10.\underline{\quad}59$. This gave me the missing 7 for 6.178. Then 3 was the only option for the blank in $-10.\underline{\quad}59$.

Partner's
Initials SS

11.1.6 Your Turn!

Component Overview: Students will find the sum for each rational number expression.



11.1.6 Your Turn!

1. Find the sum for each expression.

a. $-1\frac{1}{2} + \left(-\frac{1}{8}\right)$

$-1\frac{5}{8}$

b. $-12.37 + 18.161$

5.791

c. $-\frac{2}{5} + \left(-7\frac{1}{4}\right)$

$-7\frac{13}{20}$

d. $-4.48 + 9.366$

4.886

e. $-233.24 + 25.8$

-207.44

f. $-4\frac{1}{7} + 17\frac{2}{9}$

$13\frac{5}{63}$



As you circulate or review student work look for common errors students are making. Consider whether that error is based on their understanding of adding signed numbers, fractions, mixed numbers, or place value. Then determine any necessary small group intervention. Fluency with rational numbers is an important focus in Grade 6 Accelerated. Students will continue to build on their current understanding throughout the year.



Provide students with blank place value charts. Allow students to fill in the charts with the given decimal numbers in order to assist them with lining up decimals and the place values.



For students who need additional support with either fractions or decimals, provide On-Ramp video and practice tools to supplement their learning.

11.1.7 Wrap-Up: Ticket Out the Door

Component Overview: Fractions and mixed numbers are given in two rows. Students will choose one number from each row with the intent to find a negative answer and will find the sum. Students should not use a calculator to complete the Wrap-Up.



11.1.7 Wrap-Up: Ticket Out the Door

- Choose a value from Row A and a value from Row B that, when added, will result in a negative answer. Then, calculate the sum.

Answers will vary.

Row A

$-1\frac{3}{5}$	$-\frac{1}{8}$	$-2\frac{1}{4}$
-----------------	----------------	-----------------

Row B

$1\frac{1}{2}$	$2\frac{6}{25}$	$\frac{19}{40}$
----------------	-----------------	-----------------

Row A

$-2\frac{1}{4}$

+

Row B

$1\frac{1}{2}$

=

Sum

$-\frac{3}{4}$



Monitor for the approach that each student takes, whether they find common denominators first or round and estimate based on approximations.



If time permits, challenge students to determine how many different possible sums they could find and order their approximations from least to greatest.

11.2 Subtracting Rational Numbers

Lesson Introduction

 Benchmark	MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.		 Learning Target	I can fluently subtract rational numbers.
 Benchmark Coherence	Building On		Working Toward	
	MA.6.NSO.2.1 Multiply and divide positive multi-digit numbers with decimals to the thousandths, including using a standard algorithm with procedural fluency. MA.6.NSO.2.2 Extend previous understanding of multiplication and division to compute products and quotients of positive fractions by positive fractions, including mixed numbers, with procedural fluency.		MA.8.NSO.1.7 Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers including exponents and radicals.	
 Essential Questions	- How can number lines be used to subtract rational numbers? - How does the process of subtracting integers apply to subtracting rational numbers?			
 Lesson Narrative	Lesson 2 demonstrates how to find the difference between two numbers by using a variety of strategies, including a number line. Students consider the impact on order in subtraction expressions.			
 Component Summary	11.2.1 Warm-Up (~5 min.)	Analysis of student reasoning for greater fraction (<i>SE p. 171</i>)	11.2.4 Guided Instruction (5 min.)	Exploring the impact of the order when finding the difference between two values (<i>SE p. 174</i>)
	11.2.2 Guided Instruction (5-10 min.)	Finding the difference between two values using number lines (<i>SE p. 172</i>)	11.2.5 Collaboration (10-15 min.)	Finding the difference between two values (<i>SE p. 175</i>)
	11.2.3 Collaboration (10-15 min.)	Using number lines to find the difference between two values (<i>SE p. 173</i>)	11.2.6 Your Turn! (5-10 min.)	Practice finding the difference between two values (<i>SE p. 175</i>)
	11.2.7 Wrap-Up (~5 min.)	Self-reflection on lesson mistake (<i>SE p. 176</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - N/A <u>Additional Terms:</u> - N/A		(Optional) Number Lines	Students may benefit from having number lines that are larger than the ones in the lesson.

Teacher Tips

Common Misconceptions

- Students may misunderstand that the value of the numbers to the left of the zero on a number line are decreasing even though the digits are increasing.

Things to Consider

- For 11.2.3 Collaboration, assist students with making the connection between the direction used in the process and the sign used in the answers.
- If time is an issue, consider assigning 11.2.6 Your Turn! as homework.

Literacy and Language Routines

Compare and Connect

This lesson provides various strategies for the students to use to solve the problems. Prior to completing the 11.2.5 Collaboration, encourage students to compare and connect the different strategies. Allow them to review the strategies and examples from the lesson so that they have a strong understanding of how to execute each strategy before they work together to solve problems.

Mathematical Thinking and Reasoning (MTR) Standard

MA.K12.MTR.3.1: Mathematical Fluency

Mathematical Fluency covers selecting effective and applicable methods for solving problems. By presenting students with a variety of strategies and processes, this lesson allows students to choose effective procedures to fluently subtract rational numbers.



Differentiate Instruction

This lesson has differentiation built in throughout each section. Students use number lines to model subtraction of two rational numbers, providing a visual entry point for learners to engage with the content. Encourage students to use the visual of the number line as needed throughout this lesson to provide an additional support as they build conceptual understanding of adding and subtracting rational numbers.

Additional differentiation opportunities are noted throughout the lesson guide.

Lesson Notes:

11.2.1 Warm-Up: Math Talk

Component Overview: Two fractions and four student discussions about which fraction is greater are presented. Students will explain which student's reasoning they agree with and will explain why they disagreed with one student's reasoning.



11.2.1 Warm-Up: Math Talk

Several students discussed which fraction was greater, $\frac{13}{19}$ or $\frac{2}{3}$. A summary of the discussion is shown.

Amy:	I changed $\frac{13}{19}$ to $\frac{13}{18}$, and since $\frac{2}{3}$ is equal to $\frac{12}{18}$, I'm about 70% confident that $\frac{13}{19}$ is greater than $\frac{2}{3}$.
Brandon:	The fractions were small enough that I could do mental math and change them into common denominators, so $\frac{13}{19} = \frac{39}{57}$, and $\frac{2}{3} = \frac{38}{57}$. Therefore, $\frac{13}{19}$ is greater than $\frac{2}{3}$.
Carlos:	I changed $\frac{13}{19}$ to $\frac{14}{20}$. Then I thought of a pizza, and that's a lot of little pieces. So, I'm about 70% confident that $\frac{2}{3}$ is bigger than $\frac{13}{19}$.
Dimitri:	I know $\frac{2}{3}$ is $\frac{12}{18}$ if I multiply the top and bottom by 6. $\frac{12}{18}$ is bigger than $\frac{13}{19}$. I'm 90% confident.



The Warm-Up provides an extensive amount of information and a variety of fractions. In order to assist the students with comprehending all of the given information, provide an alternative way to organize the given information such as a three-column table:

Name	Discussion	Answer

- Explain which student's reasoning you agree with.
Answers will vary. I agree with Amy because she compared common denominators. I agree with Brandon because he used the correct process. I agree with Carlos because he chose the bigger numbers. I agree with Dimitri because a smaller numerator could be a bigger fraction.
- Choose one student you didn't agree with, and explain why you did not agree.
Answers will vary. I disagree with Amy because she did not convert correctly. I disagree with Brandon because it was more difficult for me to use mental math like he did. I disagree with Carlos because he did not use a math process to get his answer. I disagree with Dimitri because he did not compare common denominators.

11.2.2 Guided Instruction: Understanding Differences

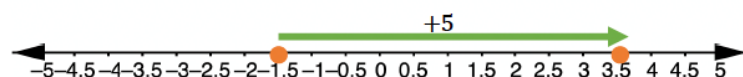
Component Overview: Students will find the difference between two rational numbers using number lines and will complete statements explaining the process.



11.2.2 Guided Instruction: Understanding Differences

Finding the **difference** of two numbers means to *subtract the numbers*. When finding the difference between two numbers, subtract the numbers in the order they are named. For example, the difference of 3.5 and -1.5 is represented by the subtraction expression $3.5 - (-1.5)$.

The difference of two numbers is how far apart the numbers are on a number line.



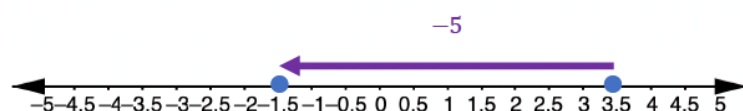
- The numbers 3.5 and -1.5 are 5 units apart. The direction to move from the second number (-1.5) to the first number (3.5) is to the ☐ left, ☒ right, or the ☐ positive ☒ negative direction.

Therefore $3.5 - (-1.5) = +5$.

- Suppose the numbers are subtracted in the opposite order, $(-1.5) - 3.5$.

The numbers 3.5 and -1.5 are still 5 units apart, but the direction of the move has changed. To move from the second number (3.5) to the first number (-1.5) is to move to the ☒ left ☐ right or the ☐ positive ☒ negative direction.

Therefore, $(-1.5) - 3.5 = -5$.



The lesson does not focus on a process. Instead, it is meant to build so the process makes sense for students. Do not reveal any rules until 11.2.4 Guided Instruction.

Students may state that bigger digits actually mean smaller values for the numbers on the left of the zero. Students may also point out that the numbers get smaller going to the left of zero, and bigger going to the right of zero.



In 11.2.2 Guided Instruction, students are provided with number lines to subtract rational numbers. By allowing the students to problem solve with number lines, students are able to visually observe the concept as well as the process for problem solving.

11.2.3 Collaboration: Understanding Differences

Component Overview: With a partner, students will write subtraction expressions and will represent those expressions on number lines. Students will also practice finding the difference between values and will note the direction used to find those differences.



11.2.3 Collaboration: Understanding Differences

- Working with your partner, write each phrase as a subtraction expression. Then, draw a number line representation to find the value of the expression. The first phrase has been done for you.

Phrase	Subtraction Expression	Number Line Representation
The difference of $-\frac{1}{5}$ and $-\frac{4}{5}$	$-\frac{1}{5} - (-\frac{4}{5})$	
Value = $\frac{3}{5}$		

Phrase	Subtraction Expression	Number Line Representation
The difference of $-\frac{4}{5}$ and $-\frac{1}{5}$	$-\frac{4}{5} - (-\frac{1}{5})$	
Value = $-\frac{3}{5}$		

Phrase	Subtraction Expression	Number Line Representation
The difference of 0.9 and -0.5	$0.9 - -0.5$	
Value = <u>1.4</u>		



Does order matter when translating from verbal to algebraic descriptions? Why or why not?



Monitor for common misconceptions.

- Students may reverse the order of terms.

$$-\frac{1}{5} - (-\frac{4}{5}) = \frac{3}{5}$$

- Students may subtract the second term as positive.

$$-\frac{4}{5} - \frac{1}{5} = -\frac{5}{5}$$

If students make one of these errors, play “My Favorite Mistake” to encourage a growth mindset.

11.2.3 Collaboration: Understanding Differences (continued)

Phrase	Subtraction Expression	Number Line Representation
The difference of -0.5 and 0.9	$-0.5 - 0.9$	
Value = -1.4		

2. With your partner, complete the table to find the difference of the values.

Expression	Distance Between the Values	Direction from Second Value to First Value	Difference
$9.9 - 6.2$	3.7	Right/ positive	$9.9 - 6.2 = 3.7$
$9.9 - (-6.2)$	16.1	Right/ positive	$9.9 - (-6.2) = 16.1$
$(-9.9) - 6.2$	16.1	Left/ negative	$(-9.9) - 6.2 = -16.1$
$(-9.9) - (-6.2)$	3.7	Left/ negative	$(-9.9) - (-6.2) = -3.7$
$6.2 - 9.9$	3.7	Left/ negative	$6.2 - 9.9 = -3.7$
$6.2 - (-9.9)$	16.1	Right/ positive	$6.2 - (-9.9) = 16.1$
$(-6.2) - 9.9$	16.1	Left/ negative	$(-6.2) - 9.9 = -16.1$
$(-6.2) - (-9.9)$	3.7	Right/ positive	$(-6.2) - (-9.9) = 3.7$



For question 2, the students must identify the correct direction used to reach the answer. Students must also record the correct positive or negative sign on each answer. Pose questions so that students make the connection between the direction used in the process and the sign of the answers, such as:

- When the direction from the second value to the first is left, what is the sign?
- When the direction from the second value to the first is positive, what is the sign?
- Would the association between the direction in the process and the sign of the answer change if the values were on opposite sides of zero?

11.2.4 Guided Instruction: Finding Differences

Component Overview: Students summarize what they learned with number lines to understanding why subtracting a negative value results in plus a positive.



11.2.4 Guided Instruction: Finding Differences

Adding a negative value has the same result as subtracting a positive value. This is because addition and subtraction have an inverse relationship.

For example, $5 + (-6)$ is equivalent to $5 - 6$. Both expressions have a value of -1 .

This relationship can be used to subtract rational numbers. The difference of two numbers can be rewritten as an equivalent addition expression. Then, the rules for addition of two numbers can be followed.

1. Fill in the blanks to complete the table.

Original Expression	$1\frac{2}{5} - \left(-\frac{6}{7}\right)$
Rewrite as an equivalent addition problem by keeping the first term, changing subtraction to addition, and changing the second term to its opposite.	$1\frac{2}{5} \quad \boxed{+} \quad \left(\boxed{+} \frac{6}{7}\right)$
Solve the problem using the rules of addition.	$\boxed{2\frac{9}{35}}$



In order to assist students with correctly applying the rules for subtraction, provide an anchor chart of rational number addition rules and ask students questions about the processes, such as:

- What is the process when the signs are the same?
- What is the process when the signs are different?

11.2.5 Collaboration: Does the Order Matter?

Component Overview: With a partner, students will evaluate subtraction expressions and explain the relationship between two expressions where the order was changed.



11.2.5 Collaboration: Does the Order Matter?

1. With your partner, determine who will be Partner A and who will be Partner B. Partner A will work the problems in Column A while Partner B works the problems in Column B. Then, work together to answer the questions that follow.

Column A	Value	Column B	Value
$3 - 2$	1	$2 - 3$	-1
$5 - (-9)$	14	$(-9) - 5$	-14
$(-11) - 2$	-13	$2 - (-11)$	13
$(-\frac{1}{6}) - (-\frac{1}{3})$	$\frac{1}{6}$	$(-\frac{1}{3}) - (-\frac{1}{6})$	$-\frac{1}{6}$
$(-1.2) - (-3.6)$	2.4	$(-3.6) - (-1.2)$	-2.4
$(-2\frac{1}{2}) - (-3\frac{1}{2})$	1	$(-3\frac{1}{2}) - (-2\frac{1}{2})$	-1

2. Explain the difference between the expressions in Column A and the expressions in Column B.
Answers will vary. The difference in the expressions is that the numbers are written in the reverse order of each other.

3. Explain the relationship between the values of the expressions in Column A and the value of the expressions in Column B.
Answers will vary. The relationship between the values is that they are opposites.



As you circulate note where students determine the pattern in the answers. If students saw the pattern and skipped determining the values of the last three expressions, then determine if they did that because they did not want to find those values or if they understood the concept. Ask students questions such as: How can you use a number line to explain what is happening between the two columns?

11.2.6 Your Turn!

Component Overview: Students work independently to find the difference between two rational numbers.



11.2.6 Your Turn!

1. What is the difference of $\frac{9}{14}$ and $-\frac{13}{20}$?

$$1\frac{41}{140}$$

2. Find the difference for each expression.

a. $5\frac{7}{8} - (-2\frac{2}{5})$

$$8\frac{11}{40}$$

b. $-45.6 - (-1.772)$

$$-43.828$$



Monitor for students who may add or subtract incorrectly. For example:

- Students may reverse the order of terms.

$$-\frac{13}{20} - \frac{9}{14} = -\frac{181}{140}$$

- Students may subtract the second term as positive.

$$\frac{9}{14} - \frac{13}{20} = -\frac{1}{140}$$

If students make one of these errors, play “My Favorite Mistake” to encourage a growth mindset.



Allow the students to associate one subtraction expression or answer with their own appropriate real-life example, such as:

- Ken stood at $-\frac{13}{20}$ feet behind the starting line, jumped $1\frac{41}{140}$ feet over the starting line, and landed $\frac{9}{14}$ feet behind the starting line.

11.2.7 Wrap-Up: Growth Mindset

Component Overview: Students will complete a self-reflection on a mistake they made in the lesson and how they can learn from it.



11.2.7 Wrap-Up: Growth Mindset

Making mistakes is one of the best ways to learn. Reflecting on our mistakes helps our brains to look for that mistake again and to avoid it the next time.

1. What mistake did you make today that you can learn from?

Means

I

Start

To

Acquire

Knowledge

Experience

Skills

Answers will vary. The mistake I made today is that I sometimes moved in the wrong direction on the number line when subtracting rational numbers.



Allow the students to elaborate about their mistakes. Provide sentence stems, such as:

I made a mistake when I _____ because _____.

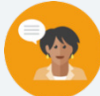
I should have _____ instead of _____.

The next time I see _____, I will _____.

11.3 Multiplying Rational Numbers

Lesson Introduction

 Benchmark	MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.		 Learning Target	I can fluently multiply rational numbers.
 Benchmark Coherence	Building On		Working Toward	
	MA.6.NSO.2.1 Multiply and divide positive multi-digit numbers with decimals to the thousandths, including using a standard algorithm with procedural fluency. MA.6.NSO.2.2 Extend previous understanding of multiplication and division to compute products and quotients of positive fractions by positive fractions, including mixed numbers, with procedural fluency.		MA.8.NSO.1.7 Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers including exponents and radicals.	
 Essential Questions	How does the process of multiplying integers apply to multiplying rational numbers?			
 Lesson Narrative	Students will also practice solving multiplication expressions and will determine if the statements will remain true for all rational numbers. Students use a calculator to determine the solutions to these expressions. Then, students will find the product of two rational numbers and will work with a partner to agree on one solution for each of the products. The lesson wraps up with two rational number expressions and three fill-in-the-blank statements about the products of two factors.			
 Component Summary	11.3.1 Warm-Up (~10 min.)	Write and solve integer equations (<i>SE p. 177</i>)	11.3.3 Guided Practice (10-15 min.)	Find the product of two rational numbers (<i>SE p. 178</i>)
	11.3.2 Collaboration (15-20 min.)	Simplify rational number expressions and complete statements (<i>SE p. 177</i>)	11.3.4 Wrap-Up (~5 min.)	Find the product of rational number expressions and complete statements (<i>SE p. 179</i>)
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> Factors		- Calculator for 11.3.2 Collaboration (Optional) Integer chips (Optional) Colored pencils	An anchor chart showing operation rules for rational numbers may be useful.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may assume that multiplying two negative numbers will result in a negative product. - Students may confuse the different processes to multiply mixed numbers, fractions, decimals, and integers.	11.3.1 Warm-Up may take some students a while to complete, especially if they are using a “guess and check” method. - Students should only use a calculator for the verification in 11.3.2 Collaboration. Consider how to monitor your students. Options could be that they do not receive a calculator until they can show you, they did the work or to come back as a whole group for the verification.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	True or False Many statements are presented in the 11.3.2 Collaboration. Consider using a True or False routine once students have filled in the blanks to complete the statements. Post statements that have different answers for the blanks and have students justify which statements are true or false.	MA.K12.MTR.3.1: Mathematical Fluency Mathematical Fluency requires flexibility and precision while carrying out calculations. By providing students with multiple opportunities to execute procedures and perform calculations, this lesson allows students to maintain flexibility while accurately multiplying rational numbers.
 Differentiate Instruction	Consider allowing students access to the 6 th Grade On-Ramp video titled Multiplying Decimals for additional support.	
Lesson Notes:		

11.3.1 Warm-Up: Bell Work

Component Overview: Students will arrange integers into equations in order to meet certain criteria. Students will add and subtract positive and negative integers and find the solutions to the equations.



11.3.1 Warm-Up: Bell Work

- Using the digits 1 to 6 at most one time each, fill in the boxes so that the top two equations are equal, and the bottom equation has the greatest value.

$$- \boxed{6} + \boxed{3} =$$

$$- \boxed{1} - \boxed{2} =$$

$$- \boxed{4} - \boxed{5} =$$



The Bell Work can inform you about which students understand operations with integers. Although Lessons 1 and 2 were focused on extending operations with integers to rational numbers this open-ended question requires more cognitive demand on student understanding. Encourage students to leave their thinking process as it could help you consider where a student is struggling.

11.3.2 Collaboration: Signed Number Rules

Component Overview: Students work collaboratively to complete statements that describe the rules for multiplying integers and discuss with a partner whether the rules apply to rational numbers. Students then multiply rational numbers and check their work using a calculator.



11.3.2 Collaboration: Signed Number Rules

- Fill in the blank with the words *positive* or *negative* to describe the rules for multiplying integers.
 - The product of two positive integers is a *positive* integer.
 - The product of two negative integers is a *positive* integer.
 - The product of a negative integer and a positive integer is a *negative* integer.
- Discuss with your partner whether you think the same rules apply to rational numbers. Write a summary of your reasoning.
Answers will vary. I think the same rules apply because integers are rational numbers.
- Complete the table. Use a calculator to find the product of each expression to verify if the same rules for signs apply when multiplying rational numbers.

Expression	Sign of Product Using the Rules	Product	Did the Same Rule Apply?
$(-\frac{3}{5})(-\frac{3}{11})$	+	$\frac{9}{55}$	yes
$\frac{7}{18}(-4\frac{8}{25})$	-	$-1\frac{17}{25}$	yes
$21.285 \cdot -3.8$	-	-80.883	yes
$(-91.09)(-6.4)$	+	582.976	yes

- Complete the sentence.
 The rules for multiplying rational numbers are

☐ the same as
☐ different from

 the integer rules for multiplication.



Students may not recall the definitions of each vocabulary word in these statements. Provide numerical examples along with each statement so that the students can connect each concept to a clear numerical representation of the statement.



Lessons 1 and 2 showed students that the rules of adding and subtracting integers applies to rational numbers. Students may base their response on the fact that it worked in the last two lessons. Consider asking these students if an integer is considered a rational number. It may also be helpful for students to understand the connection of the types of numbers from whole numbers to rational numbers.



True or False

Once students have completed the statements from questions 1 and 4, engage students in a True or False routine to synthesize the information presented in this section. Display statements with different answers and have students discuss and explain which statements are true/false and how they know.

11.3.3 Guided Practice: Applying Signed Number Rules to Rational Numbers

Component Overview: Students work with a partner to multiply rational numbers in the table. Each student works in their respective column and then checks their product with their partner to resolve any differences.



11.3.3 Guided Practice: Applying Signed Number Rules to Rational Numbers

1. With your partner, decide who will be Partner A and who will be Partner B. Then, find the product of the expression in the column that matches your letter.

Column A	Product	Column B
$790 \left(\frac{1}{10}\right)$	79	$7.9(10)$
$\left(-\frac{6}{7}\right) \cdot 7$	-6	$(0.1) \cdot (-60)$
$2.1 \cdot (-2)$	-4.2	$(-8.4)\left(\frac{1}{2}\right)$
$2.5 \cdot -3.25$	-8.125	$\left(-\frac{5}{2}\right)\left(\frac{13}{4}\right)$

The expressions in each row have the same product. Check your work with your partner, and if necessary, resolve any differences. Write the agreed-upon product in the center column.



During the Guided Practice, students must find their own answers and then work with their partners to agree on one solution. Prior to completing the table, discuss strategies students can use to help them come to an agreement on one solution.

Students may suggest that each partner solves all questions, and once both partners reach the same answer, they choose that answer as the solution.



Offer students the option to convert each answer into an equivalent form. Some students may feel more comfortable working with fractions or with decimals, so an opportunity for differentiation is to allow them to convert to the equivalent rational numbers of their choice.

11.3.3 Guided Practice: Applying Signed Number Rules to Rational Numbers (continued)

2. Choose a value from Row A and a value from Row B that, when multiplied, will result in a negative answer. Then, calculate the product.

Row A

$-9\frac{2}{7}$	$-\frac{3}{5}$	$12\frac{2}{9}$
-----------------	----------------	-----------------

Row B

$4\frac{3}{8}$	$\frac{11}{16}$	$-\frac{9}{50}$
----------------	-----------------	-----------------

Row A

×

Row B

=

Product

$$-9\frac{2}{7}$$

$$-9\frac{2}{7}$$

$$-\frac{3}{5}$$

$$-\frac{3}{5}$$

$$12\frac{2}{9}$$

$$4\frac{3}{8}$$

$$\frac{11}{16}$$

$$4\frac{3}{8}$$

$$\frac{11}{16}$$

$$-\frac{9}{50}$$

$$-40\frac{5}{8}$$

$$-6\frac{43}{112}$$

$$-2\frac{5}{8}$$

$$-\frac{33}{80}$$

$$-2.2$$



For students who need additional support with either fractions or decimals, provide On-Ramp video and practice tools to supplement their learning.



If time permits, challenge students to determine how many different possible sums they can find and order their approximations from least to greatest.

11.3.4 Wrap-Up: Ticket Out the Door

Component Overview: Students will solve two rational number expressions and will fill in the blanks for three statements about the products of two factors. Students should not use a calculator to complete the Wrap-Up.



11.3.4 Wrap-Up: Ticket Out the Door

- Find the product of each expression.

a. $(0.24)(-3.8)$
 -0.912

b. $\left(-\frac{2}{5}\right)\left(-\frac{9}{8}\right)$
 $\frac{9}{20}$

- Fill in the blanks with the words *positive* or *negative*.

- Two positive factors result in a positive product.
- Two negative factors result in a positive product.
- A positive factor and a negative factor result in a negative product.

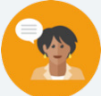


Allowing the students to solidify their responses with examples of each statement is an opportunity to extend the lesson. Encourage the students to come up with their own numerical equations that would support each statement.

11.4 Dividing Rational Numbers

Lesson Introduction

Lesson Introduction						
 Benchmark	MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.		 Learning Target	I can fluently divide rational numbers.		
 Benchmark Coherence	Building On		Working Toward			
	MA.6.NSO.2.1 Multiply and divide positive multi-digit numbers with decimals to the thousandths, including using a standard algorithm with procedural fluency. MA.6.NSO.2.2 Extend previous understanding of multiplication and division to compute products and quotients of positive fractions by positive fractions, including mixed numbers, with procedural fluency.		MA.8.NSO.1.7 Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers including exponents and radicals.			
 Essential Questions	- How does the process of dividing integers apply to dividing rational numbers? - How do I divide positive and negative rational numbers?					
 Lesson Narrative	This lesson builds on the integer rules for division as students review the rules and apply them to the division rational numbers. A brief review of the division of positive rational numbers and the sign rules for the division of integers are used to help students determine the sign rules for the division of rational numbers. Students apply the sign rules to divide rational numbers.					
 Component Summary	11.4.1 Warm-Up (~5 min.)	Determine the value of a stack of quarters (<i>SE p. 180</i>)	11.4.4 Guided Practice (10 min.)	Apply the sign rules to the division of positive and negative rational numbers (<i>SE p. 183</i>)		
	11.4.2 Refresher (10 min.)	Review the division of positive rational numbers (<i>SE p. 181</i>)	11.4.5 Your Turn! (10 min.)	Practice division of positive and negative rational numbers (<i>SE p. 183</i>)		
	11.4.3 Collaboration (10 min.)	Review sign rules for the division of integers (<i>SE p. 182</i>)	11.4.6 Wrap-Up (~5 min.)	Lesson reflection (<i>SE p. 184</i>)		
 Resources	Glossary		Materials		Additional Preparation	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		(optional) Calculators (to check work)		N/A	

 Teacher Tips	Common Misconceptions	Things to Consider															
	- Students may change the divisor to its opposite rather than the reciprocal when dividing. - Students may incorrectly change an improper fraction to a mixed number, forgetting that any remainders need to be written as a fraction with the given denominator.	- If students ask why the rules for multiplying and dividing signed numbers are the same it may be helpful for them to be reminded that division such as $\frac{2}{3} \div -4$ is the same as $\frac{2}{3} \cdot -\frac{1}{4}$. - Students should only use a calculator for the verification in 11.4.3 Collaboration. Consider how to monitor your students. Options could be that they do not receive a calculator until they can show you, they did the work or to come back as a whole group for the verification. - Determine if your students need the support of all or part of the refresher.															
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard															
	Notice and Wonder 11.4.4 Guided Practice positions students to apply the sign rules for dividing rational numbers without actually solving the given expressions. Consider applying “Notice and Wonder” to give students who may still be struggling with sign rules an opportunity to talk through the task to solidify their thinking before moving on to 11.4.5 Your Turn!	MA.K12.MTR.3.1: Mathematical Fluency Students will be applying the sign rules for division to divide rational numbers fluently. Clarifications within this standard emphasize “Adapt(ing) procedures to apply them to a new context” – utilize the progression of the lesson from reviewing the division of positive rational numbers and the rules for dividing integers to bringing them together to determine how to divide positive and negative rational numbers in 11.4.4 Guided Practice.															
 Differentiate Instruction	The lesson’s objective is for students to divide rational numbers fluently. In 11.4.3 Collaboration, partners work together to review the sign rules for the division of integers. Both are brought together in 11.4.4 Guided Practice, where students divide positive and negative rational numbers. Consider using visual models for students who are still struggling to remember the rules and procedures. A three-column chart can be used to record the sign rules for division, and number lines or other visual models can help students review the division of integers and to show the division of an integer and a rational number. <table border="1" data-bbox="1654 843 2041 987"> <thead> <tr> <th>Dividend</th><th>Divisor</th><th>Quotient</th></tr> </thead> <tbody> <tr><td> </td><td> </td><td> </td></tr> <tr><td> </td><td> </td><td> </td></tr> <tr><td> </td><td> </td><td> </td></tr> <tr><td> </td><td> </td><td> </td></tr> </tbody> </table>		Dividend	Divisor	Quotient												
Dividend	Divisor	Quotient															
Lesson Notes:																	

11.4.1 Warm-Up: Would You Rather?

Component Overview: Students determine the value of a stack of quarters as tall as they are and select whether they would choose this value or \$225.00.



11.4.1 Warm-Up: Would You Rather?

1. Explain if you would rather have a stack of quarters that stands from the floor to the top of your head **OR** \$225.00.

Answers will vary. I would rather have \$225.00, if I am 62 inches or less. The width of a quarter is 0.069 inches.

$$\frac{62}{0.069} \approx 898.55 \cdot 0.25 = 224.64$$

\$224.64

2. Explain whether you would change your mind if the quarters were stacked to the top of your desk.

Answers will vary. No, I am taller than my desk, so I would still rather have \$225.00.



Consider engaging prior knowledge on division by prompting students with these questions:

- If a quarter is about 0.069 of an inch thick, how can you use the thickness to estimate which stack you would rather have?
- Which operation makes the most sense to help with estimation? Why?

11.4.2 Refresher: Dividing Positive Rational Numbers

Component Overview: Students review the division of positive rational numbers; in particular, a decimal divided by another decimal and a fraction divided by a fraction.



11.4.2 Refresher: Dividing Positive Rational Numbers

- Ren found the following quotient, making at least one error in her work.
 - Circle the error(s) in Ren's work and write an explanation of the error(s). Then, find the correct quotient.

Ren's Work	Explanation	Correct Quotient
$12.636 \div 3.12$ 	<i>Ren should have moved the decimal point two times in the divisor and dividend – making the divisor a whole number.</i>	$12.636 \div 3.12$

- What advice do you have for Ren that could help her with dividing decimals in the future?
Answers will vary. One thing to remember when making your divisor a whole number is moving the decimal point in the dividend the same number of times as the divisor.

- Trevor found the following quotient, making at least one error in his work. Circle the error(s) in Trevor's work. Write an explanation of the error(s). Then, find the correct quotient.

Trevor's Work	Explanation	Correct Quotient
$\left(\frac{38}{13}\right) \div \left(1\frac{6}{13}\right)$ $\left(\frac{38}{13}\right) \div \left(\frac{19}{13}\right)$ 	<i>Answers will vary. Trevor should have applied the multiplicative inverse to find quotient.</i>	$\frac{38}{13} \div \frac{19}{13}$



Encourage students to use precise mathematical vocabulary by prompting them to use the terms 'divisor' and 'dividend' in their response to question 1b.



Some students may struggle with keeping the numbers and decimal points in alignment when using long division. Consider having them rewrite and solve the question using grid paper to work it out, and then copy it into the workbook or cutting the grid paper to fit and glue or tape into the workbook.



Students may mention that Trevor should have multiplied by the reciprocal. Consider asking students to clarify their answers:

- The reciprocal of what -- the dividend, the divisor, or both?
- What property of equality is used when multiplying by the reciprocal?

11.4.3 Collaboration: Signed Number Rules

Component Overview: Students review the sign rules for the division of integers. Students determine that same sign rules are also applied when working with the division of rational numbers.



11.4.3 Collaboration: Signed Number Rules

- Fill in the blanks with the words *positive* or *negative* to write the rules for dividing integers.
 - The quotient of two positive integers is a positive integer.
 - The quotient of two negative integers is a positive integer.
 - The quotient of a negative integer divided by a positive integer is a negative integer.
 - The quotient of a positive integer divided by a negative integer is a negative integer.
- Discuss with your partner whether you think the same rules apply to rational numbers. Write a summary of your reasoning.
Answers will vary. The rules for dividing rational numbers is the same as the rules for dividing integers. This is true because division can be rewritten as multiplication by the multiplicative inverse of the divisor.
- Complete the table. Use a calculator to find the quotient of each expression to verify if the same rules for signs apply when dividing rational numbers.

Expression	Sign of Quotient Using the Rules	Quotient	Did the Same Rule Apply?
$(-2\frac{3}{8}) \div (-1\frac{1}{4})$	Positive	$1\frac{9}{10}$	Yes
$6\frac{7}{18} \div -4\frac{2}{9}$	Negative	$-1\frac{39}{76}$	Yes
$43.008 \div -4.096$	Negative	-10.5	Yes
$(-525.6) \div (-0.36)$	Positive	$1,460$	Yes

- Complete the sentence.
The rules for dividing rational numbers are

- ☐ the same as
☐ different from

the integer rules for division.



Some students may struggle to keep track of the sign rules. Consider having them build a three-column chart first, with the labels 'Dividend,' 'Divisor,' and 'Quotient'. Have them fill in the columns with either the signs only or using actual positive and negative numbers to summarize the four rules.



Students should only use a calculator for the verification of the answer so that students can formally state the rule.

Consider asking students if the rules also apply if the forms of the rational numbers differ as in $-4.25 \div \frac{1}{3}$.



Have students write and solve their own expressions where they are dividing a fraction by a decimal and a decimal by a fraction. Challenge them further by asking them to write one of the fractions with a decimal numerator and denominator.

11.4.4 Guided Practice: Finding the Quotients of Rational Numbers

Component Overview: Students apply the sign rules for the division of rational numbers. Students write and solve their own division sentences using rational numbers.



11.4.4 Guided Practice: Finding the Quotients of Rational Numbers

1. Circle the expressions that will result in a positive quotient.

$$-9.655 \div 0.5$$

$$0.068 \div -0.01$$

$$-14.628 \div -4.6$$

$$\frac{16}{20} \div \left(-\frac{8}{15}\right)$$

$$\left(-\frac{9}{6}\right) \div \left(-\frac{5}{3}\right)$$

$$\left(-\frac{3}{5}\right) \div 2\frac{1}{2}$$



Notice and Wonder

Before students start to circle the expressions in question 1, consider utilizing a “Notice and Wonder” routine to offer students who still are struggling with sign rules an opportunity to talk through the task to solidify their thinking.

- What do you notice? What is the same? What is different?
- What do you wonder? Can you determine the sign of the quotient without actually solving? How?

- 2.
- a. Write a division sentence in the form of $a \div b$ where the value of a is a number **between** 15.1 and 16.2, and b is a number **between** -4.6 and -3.3 .

Answers will vary.

$$15.3 \div -4.3 = -3.558$$

- b. What is the value of the quotient?

Answers will vary.

$$15.3 \div -4.3 = -3.558$$

11.4.5 Your Turn!

Component Overview: Students find the values of given expressions using rational numbers.



11.4.5 Your Turn!

1. Find the value of each expression.

a. $\left(-3\frac{5}{9}\right) \div \left(3\frac{1}{3}\right)$

$$\begin{aligned} -\frac{32}{9} \div \frac{10}{3} &= -\frac{32}{9} \times \frac{3}{10} \\ &= -\frac{32}{30} = -1\frac{1}{15} \end{aligned}$$

b. $70.62 \div -160.5$

-0.44

c. $\left(-1\frac{6}{7}\right) \div \left(-5\frac{3}{4}\right)$

$$\begin{aligned} -\frac{13}{7} \div -\frac{23}{4} \\ &= -\frac{13}{7} \times -\frac{4}{23} \\ &= \frac{52}{161} \end{aligned}$$

d. $-44.64 \div -2.88$

15.5

e. $8\frac{1}{5} \div \frac{3}{2}$

$$\frac{41}{5} \cdot \frac{2}{3} = \frac{82}{15} = 5\frac{7}{15}$$

f. $-45.018 \div (-7.32)$

6.15



Students may struggle to find the quotients of mixed numbers. Remind students to convert each mixed number to a fraction greater than one before multiplying. Encourage students to apply the rules of dividing integers and pay attention to the signs of the dividend and divisor.



Consider providing On-Ramp to 6th Grade videos and practice for students who need additional scaffolding.

- Dividing Decimals
- Division of a Whole Number by a Unit Fraction

Videos are available in Spanish and English.

11.4.6 Wrap-Up: Reflection

Component Overview: Students reflect on their learning for today's lesson.



11.4.6 Wrap-Up: Reflection

Draw an **X** on each line to indicate where your current learning is in relation to each statement. Then, provide a brief explanation or evidence to support your self-assessment.

1. I can correctly decide the sign of the quotient when dividing rational numbers.

I need help.
I can't get started. I'm getting there. I understand and have evidence.



Evidence:

Answers will vary.

2. I can fluently divide rational numbers that are fractions.

I need help.
I can't get started. I'm getting there. I understand and have evidence.



Evidence:

Answers will vary.

3. I can fluently divide rational numbers that are decimals.

I need help.
I can't get started. I'm getting there. I understand and have evidence.



Evidence:

Answers will vary.



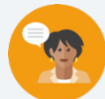
For students who respond with “I’m getting there,” consider prompting them to specifically identify where they are struggling so that it can be addressed.

Compare that with any observations you made during the lesson. What were common mistakes made by your students? What were some roadblocks your students may have had? Consider whether your students may benefit from a day of practice or small-group intervention.

11.5 Solving Real-World Problems With Rational Numbers – Part 1

Lesson Introduction

 Benchmark	MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers.		 Learning Target	I can solve problems with one or two operations by adding, subtracting, multiplying, and/or dividing rational numbers.
 Benchmark Coherence	Building On		Working Toward	
	MA.6.NSO.2.3 Solve multi-step, real-world problems involving any of the four operations with positive multi-digit decimals or positive fractions, including mixed numbers.		MA.8.NSO.1.7 Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers including exponents and radicals.	
 Essential Questions	- How do you solve multi-step problems involving addition and subtraction of rational numbers? - How do you solve multi-step real-world problems using multiplication and division of rational numbers?			
 Lesson Narrative	Lesson 5 opens a series of lessons on solving problems using any of the four operations in a variety of real-world contexts. Students focus on problems involving just one or two operations before extending to more complex scenarios in Lessons 6 and 7.			
 Component Summary	11.5.1 Warm-Up (~5 min.)	Identify patterns in division equations (<i>SE p. 185</i>)	11.5.3 Guided Instruction (10-15 min.)	Using key information and drawing diagrams, break into steps to solve real-world, multi-step problems with rational numbers (<i>SE p. 186</i>)
	11.5.2 Collaboration (10-15 min.)	Draw a diagram to solve real-world problems with rational numbers (<i>SE p. 185</i>)	11.5.4 Your Turn! (10-15 min.)	Solve real-world, multi-step problems with rational numbers (<i>SE p. 188</i>)
	11.5.5 Wrap-Up (~5 min.)	Sometimes, always, or never growth mindset reflection (<i>SE p. 189</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Credit - Debit - Equations - Expressions		(Optional) Chart paper (Optional) Markers - Calculator	N/A

 Teacher Tips	Common Misconceptions - Students may have trouble representing values in the questions with expressions and equations. - Students may struggle to add and subtract mixed numbers with different denominators. - Students may not recognize that when asked to describe how a value changes after a decrease, a negative number should be used.	Things to Consider - Consider your students' experience with banks and checking accounts. If needed, discuss what students know about how bank accounts work, and have them read through the question to note any language they are unfamiliar with. - Allowing students to use calculators during the lesson will help students focus on the real-world contexts. Consider having no-calculator questions where students turn their calculator over. Teachers should look for and consider opportunities for students to improve their fluency with rational numbers.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Three Reads 11.5.2 Collaboration presents students with a lot of text to work through when solving the given question. Consider utilizing “Three Reads” when presenting the question, positioning students to make sense of the context, highlight numbers and quantities, and identify key information in the question before beginning to solve.	MA.K12.MTR.7.1: Real-World Contexts Students engage in real-world problem solving throughout the lesson. MTR 7 asks students to “Use models and methods to understand, represent and solve problems.” Students utilize ‘Three Reads’ in this lesson to help them make sense of the question as they read for context, to highlight numbers and quantities, and to identify key information, drawing diagrams. and breaking the questions down into simpler steps to solve.
 Differentiate Instruction	Students can engage with On-Ramp videos on the following topics for additional support: - Real-World Multi-Step Addition and Subtraction Problems - Subtract Fractions with Unlike Denominators (11.5.3 Guided Instruction question 2)	
Lesson Notes:		

11.5.1 Warm-Up: Notice and Wonder

Component Overview: Students engage in a Notice and Wonder routine to examine patterns in division equations.



11.5.1 Warm-Up: Notice and Wonder

Several division equations are shown.

$$0.28 \div 0.07 = \square$$

$$2.8 \div 0.7 = \square$$

$$28 \div 7 = 4$$

$$280 \div 70 = 4$$

$$2,800 \div 700 = 4$$

$$28,000 \div 7,000 = 4$$

1. What do you wonder about the pattern?

Answers will vary. Will the pattern continue with the first two expressions? How does the decimal point affect the solution?

2. What do you notice about the equations?

Answers will vary. The solution to all the expressions in the pattern is 4.

3. Use the pattern in the equations to determine the quotients of the first two equations.

$$0.28 \div 0.07 = 4$$

$$2.8 \div 0.7 = 4$$



Notice and Wonder

Notice and Wonder activities allow students to lead the conversation. Encourage students to notice that the decimal place is changing in each equation while the quotient remains the same. Highlight student responses that recognize this and wonder what that means for the missing quotients.

11.5.2 Collaboration: Solving a Problem with Rational Numbers

Component Overview: Students work together to solve a real-world, multi-step problem with rational numbers. Students draw a diagram to model the problem and use the diagram to help answer questions about the problem.

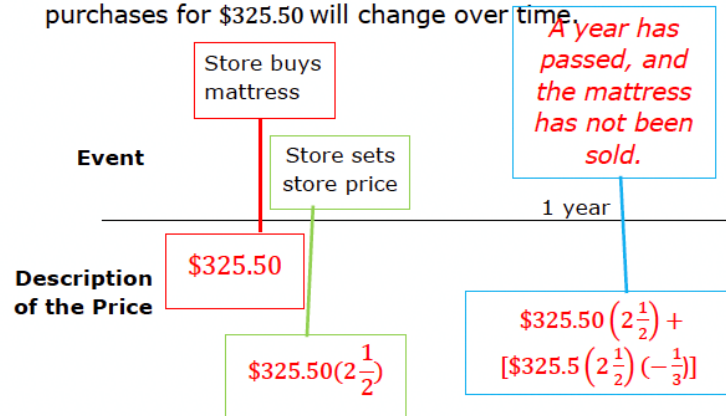


11.5.2 Collaboration: Solving a Problem with Rational Numbers

In order to make a profit, stores sell products for more than what they paid for them. For example, a local furniture store buys a mattress from a manufacturer for a certain price. The store then sets the store price at $2\frac{1}{2}$ times what they paid the manufacturer for it. If the mattress hasn't sold in a year, then the store will reduce its price by $\frac{1}{3}$ of the original store price.

Work with your partner to complete the following based on the information given.

1. Drawing a diagram can help make sense of the problem. Without calculating, complete the timeline to show how the price of a mattress the store purchases for \$325.50 will change over time.



Three Reads

Consider engaging students in 'Three Reads' to prepare them to make sense of the question.

Begin by reading the question aloud to the whole group after posing the prompt, "What is this situation about?" Students sit with their partner and discuss, sharing with the whole group when finished.

Pose the second prompt, "What are the quantities in this situation?" Have partners read through the second time and bring the whole group together again; list quantities identified on chart paper or whiteboard.

Pose the third prompt, "What mathematical questions can we ask about this situation?" Have partners again read through for a third time, asking them to come up with two questions. Once again, bring the whole group together to share their questions. Consider prompting students to think about the context and identified quantities by asking, "Can this question be answered using the information we've already identified in the problem?"

At this point, partners begin to collaborate to work through and answer the questions.

11.5.2 Collaboration: Solving a Problem with Rational Numbers (continued)

2. Explain whether the store's first price will be greater than or less than the price they paid the manufacturer.

Answers will vary. The store's first price will be greater than the price the local furniture store paid the manufacturer because the store marked up the price $2\frac{1}{2}$ times.

3. What is the first price the store will set for the mattress?

\$813.75

4. Explain whether the price after a year will be greater than or less than the store's first price.

Answers will vary. The price after a year will be less than the store's first price because the store reduces the price by $\frac{1}{3}$ of the original store price to increase the chance of selling the mattress.

5. What is the sale price of the mattress after one year?

\$542.50

6. If the store sells the mattress after one year, explain if the store will make or lose money on the mattress.

Answers will vary. The store will make a profit from the original purchase of the mattress for \$325.50 after the original increase and reduction in price after a year.



Refer to information recorded in 'Three Reads' routine and the timeline students filled in for question 1:

- What quantity describes how the store sets the price for the mattress?
- Which price represents what the store paid for the mattress?
- How does the price change once the store sets the price for the mattress?
- Which quantity represents the change in the value of the mattress after one year?



Encourage students to answer one or two of the identified questions in the 'Three Reads' practice at the beginning of 11.5.2 Collaboration.

11.5.3 Guided Instruction: Solving Real-World Problems with Rational Numbers

Component Overview: Students solve real-world, multi-step problems with rational numbers by pulling out key information, drawing diagrams, and breaking it down into each step.



11.5.3 Guided Instruction: Solving Real-World Problems with Rational Numbers

1. Alexis has a monthly subscription to the video streaming service ViewNow for \$7.20 per month. For her birthday she received a ViewNow gift card in the amount of \$55.00.
 - a. Write an expression using addition that represents the balance on the card after one month if the only charge on the gift card is the monthly subscription.
 $55 + (-7.20)$
 - b. Complete the table by filling in values on each blank line to record the activity on the account each month. Continue each month to determine how many months it will take before Alexis does not have enough money to pay the monthly fee using the gift card. Part of Month 1 has been completed for you.

Month	Starting Balance	Change in Account	Ending Balance
1	\$55.00	$\$55.00 + (-\$7.20)$	\$47.80
2	\$47.80	$\$47.80 + (-\$7.20)$	\$40.60
3	\$40.60	$\$40.60 + (-\$7.20)$	\$33.40
4	\$33.40	$\$33.40 + (-\$7.20)$	\$26.20
5	\$26.20	$\$26.20 + (-\$7.20)$	\$19.00
6	\$19	$\$19.00 + (-\$7.20)$	\$11.80
7	\$11.80	$\$11.80 + (-\$7.20)$	\$4.60
8			
9			

- c. How many months will it take before Alexis does not have enough money left on the gift card to pay the monthly charge?
7 Months



Students may decide the service fee is subtracting the positive number (the fee), while other students may decide to add the fee as a negative value. Both ways work, but make sure they understand not to both subtract and use a negative number to represent the fee. The number of rows given on the page is not an indication of the answer. There may be empty rows.



The tables provide a visual means for students to organize the information presented in the question, providing a visual entry point for learners. Encourage students to discuss their strategies for solving with a partner to provide additional auditory entry points.

11.5.3 Guided Instruction: Solving Real-World Problems with Rational Numbers (continued)

2. The **departure from the average** is the difference between the actual amount of rain and the average amount of rain for a given month. The departure from the average rainfall, in inches (in.), for Orlando, Florida for June, July, and August is shown in the table.

June	July	August
$-3\frac{19}{20}$ in.	$-1\frac{7}{10}$ in.	$4\frac{5}{8}$ in.

- a. If the average rainfall for June is $7\frac{29}{50}$ in., what was the actual rainfall in June?
 $3\frac{63}{100}$
- b. What is the difference in the departure from the average rainfall from July to August?
 $4\frac{5}{8} - (-1\frac{7}{10}) = 6\frac{13}{40}$
3. Ramon paid \$18,360.00 for a new car. By the end of the first year, the value of the car changed by $-\frac{3}{14}$ of its original value. By the end of the second year, the value of the car changed by $-\frac{2}{9}$ of its value at the end of the first year.
- a. Is the value of the car increasing or decreasing during these two years?
The value of the car is decreasing during the two years after purchase.
- b. By what amount did the value of the car change at the end of the first year?
 $-\$3,934.29$
- c. What is the value of Ramon's car at the end of the second year?
 $\$11,220$
4. A water well drilling rig dug to a depth of -147.5 feet (ft). The rig drills $-2\frac{1}{2}$ ft in one hour.
- a. How many hours has the drill been running?
 $-147.5 \div (-2.5) = 59 \text{ hours}$
- b. At this rate, how many hours will it take until the drill reaches a depth of -250 ft?
 $-250 \div (-2.5) = 100 \text{ hours}$



Departure from the average may be a new term for students but they have likely used the idea before when comparing values to averages. Pause to have students consider contexts where they may have used the idea themselves. One possible context is grades. If a student makes an 85 on a test but their test average is 91 then the 85 is a difference of 6 from the average.

Students may struggle to find the departure from the average. Ask guiding questions to help students subtract mixed numbers with different denominators:

- What is the least common multiple for these denominators? How can you use LCM to find the departure from the average?
- What operation can you use to find the departure from the average? Why does that work?



Provide students with some background information on depreciation. Most goods have a depreciation. Some goods may depreciate and then after time their value increases because they become antiques. Explain to students that the $-\frac{3}{14}$ means that this is the amount of the value that the car loses during its first year. Another way of stating that is that at the end of the first year the car is $\frac{11}{14}$ of its original value.

Students may not identify the value of the car after the first year decreased, so the change in value is represented by a negative number. Consider asking which students wrote a positive value and which wrote a negative value. Then, choose one student from each to explain their reasoning. If students state that the car is depreciating \$3,934.29 then the negative is inferred.

11.5.4 Your Turn!

Component Overview: Students solve real-world, multi-step problems with rational numbers.



11.5.4 Your Turn!

- The table shows the change in value for a share of stock in a company at the end of each week during the month of December. A negative value represents the amount the share decreased, and a positive value represents the amount the share increased.

December
-\$22.65
\$225.30
-\$106.42
\$91.10

If the value of the stock began at \$75.37 at the start of the month, what is the final value of the share at the end of the month?

$$\$75.37 - \$22.65 + \$225.30 - \$106.42 + \$91.10 = \$262.70$$

- Samuel is practicing for his town's annual high school quarterback football-throwing competition. Last year's champion threw the football $40\frac{1}{4}$ yards (yd). A summary of Samuel's throws in relation to the throws of last year's champion is shown. Complete the table to determine how far Samuel threw the football during each of his tries.

Throw	Comparison of Samuel's Throw to Last Year's Champion	Expression	Yards Thrown
1 st	$+3\frac{1}{3}$ yards	$40\frac{1}{4} + 3\frac{1}{3}$	$43\frac{7}{12}$
2 nd	$-\frac{9}{7}$ yards	$40\frac{1}{4} + \left(-\frac{9}{7}\right)$	$38\frac{27}{28}$
3 rd	$-2\frac{6}{13}$ yards	$40\frac{1}{4} + \left(-2\frac{6}{13}\right)$	$37\frac{41}{52}$



The contexts may not be familiar to students. Encourage students to use the information in the context to determine what to do. Placing students in new situations helps them to apply the content they have learned. Allow students to have productive struggle. Direct students with questions that may help them retrieve the information themselves.

As you circulate note whether students are struggling with operations or with the context. Use this information to inform your instruction on the remaining lessons of this section.

11.5.4 Your Turn! (continued)

3. Delaney wants to sell her bicycle. The original price of the bike was \$150. The value of the bike has changed by $-\frac{2}{5}$ of the original price. What is the current value of the bike?

\$90

4. At the beginning of the day on Monday, Stephon had \$15.78 in his lunch account. At the end of the day on Friday, he had \$1.53 in his lunch account. If Stephon purchased the same lunch each day on Monday through Friday, describe how the account changed each day.

The account decreased by \$2.85 per day.



Encourage students who may be struggling with question 4 to refer to the diagram they used in 11.5.2 Collaboration question 1.

11.5.5 Wrap-Up: Growth Mindset

Component Overview: Students will use the terms “sometimes,” “always,” and “never” to describe how each statement represents their feelings in relation to a growth mindset.



11.5.5 Wrap-Up: Growth Mindset

- As you reflect on today’s lesson, circle the word *Sometimes*, *Always*, or *Never* to indicate how often each statement describes you.

Answers will vary.

When I work hard on a problem, I expect that the answer will be correct.	Sometimes	Always	Never
There is no point in trying to do a problem if I know I won’t get the correct answer.	Sometimes	Always	Never
Even if I made a mistake, I can still learn from working through a problem.	Sometimes	Always	Never
I can show that I understand how to solve a problem, even if there is an error in my work.	Sometimes	Always	Never
If my answers are incorrect even after working extremely hard, I still choose to work hard on the next problem.	Sometimes	Always	Never



Consider prompting students to choose one of the statements and provide an example and justification for their response. Encourage them to use an example from this lesson or Unit 11.

Use student feedback to help you consider which students may need to be encouraged to embrace a growth mindset.

11.6 Solving Real-World Problems With Rational Numbers – Part 2

Lesson Introduction

 Benchmark	MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers.		 Learning Target	I can solve problems with three operations involving any of the four operations with rational numbers.
 Benchmark Coherence	Building On		Working Toward	
	MA.6.NSO.2.3 Solve multi-step real-world problems involving any of the four operations with positive multi-digit decimals or positive fractions, including mixed numbers.		MA.8.NSO.1.7 Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers including exponents and radicals.	
 Essential Questions	- How can you determine which form of rational number should be used when solving real-world problems with fractions and decimals? - How do you solve multi-step real-world problems involving any of the four operations with rational numbers?			
 Lesson Narrative	Lesson 6 extends students' comprehension of real-world problems to analyze the work of others within that context. 11.6.2 Activity drives learning outcomes in this lesson, with students actively engaging with each other and with real-world content by analyzing the written work of students provided in the Workbook. Students are not only identifying which operations are used in each context, but also evaluating the approaches, algorithms, and calculations of their peers.			
 Component Summary	11.6.1 Warm-Up (~5 min.)	Identify a math fail on a thermometer (<i>SE p. 190</i>)	11.6.3 Collaboration (10 min.)	Fill in missing information for a real-life example using operations and rational numbers (<i>SE p. 194</i>)
	11.6.2 Activity (25 min.)	Analyze examples of student work using operations and rational numbers (<i>SE p. 191</i>)	11.6.4 Wrap-Up (~5 min.)	Self-reflection over a positive aspect, progress made, and a confusing part from the lesson (<i>SE p. 195</i>)
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - N/A <u>Additional Terms:</u> - Order of operations		(Optional) Calculator	- An anchor chart showing the processes to convert equivalent rational numbers may be useful. 11.6.1 Warm-Up has students identifying a math fail on a thermometer. Consider projecting or enlarging the thermometer for students who might struggle with reading the scale.

 Teacher Tips	Common Misconceptions - Students may use the incorrect operation to find the fraction of a number. - Students may use the incorrect place value to round decimals that represent money, like using .5 for five cents or .05 for 50 cents.	Things to Consider 11.6.2 Activity includes performing operations with several different forms of rational numbers. Consider posting or reviewing the processes to convert to equivalent forms of rational numbers. - For question 4 in the 11.6.2 Activity, students may need support in understanding why they need to add the ratios first to find the number of ounces in one batch of dye before finding how much of each dye they need for 20 oz. - Consider not allowing students to use a calculator to build students fluency with operations using rational numbers. - Consider creating an observational chart to use as you circulate. Determine what aspects of the learning target, the benchmark, and MTR standards you would like to observe.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share 11.6.2 Activity provides an opportunity for students to engage in a Think-Pair-Share routine as they complete Error Analyses for four student work samples. Encourage students to consider each display of student work, share their analysis with a partner, and revise or build on their responses before putting their answers in writing. This encourages mathematical discussion and deepens conceptual understanding.	MA.K12.MTR.4.1: Discussion and Reflection Students are actively engaging in analyzing the work of others throughout this lesson, particularly in working through 11.6.2 Activity. Activities like this facilitate a “culture in which students ask questions of the teacher and their peers, and an error is an opportunity for learning,” as stated in MA.K12.MTR.4.1 clarifications. Encourage and celebrate questions and students who change or adjust their thinking based on the questions or reasoning of others.
 Differentiate Instruction	Task Cards of the student examples could be provided for the 11.6.2 Activity. - Students can engage with each task card at stations to incorporate movement and place when solving problems. For example, you could decorate each station to anchor students in the context (i.e., task card 1 could be the Six Flags station). - Instead of rotations with different task cards, students could be assigned numbered groups to work through a set of Task Cards to then find differing partners or pairs to either compare and verify answers, or to teach as a small-group instructor to their peers. - For students who struggle with calculations consider incorporation a calculator station. Each pair may be given a “ticket” which they can trade in to use the calculator.	
Lesson Notes:		

11.6.1 Warm-Up: Math Fail

Component Overview: Students identify a math fail on a thermometer that does not display negative numbers below zero.

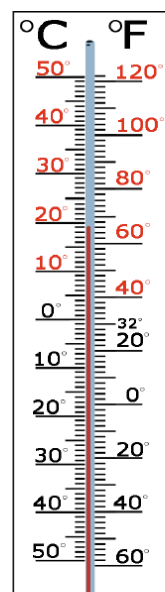


11.6.1 Warm-Up: Math Fail

A picture of a thermometer is shown.

1. Explain the error in the image.

The numbers below zero should have a negative sign. The label for 32° is in the wrong place.



For students who may struggle to read the thermometer with smaller print, consider projecting a larger image of it for students to use while answering the questions.

2. Describe how to fix the image so that there is no “math fail.”

Add a negative sign to all of the numbers below zero. 32° should be across from 0°.

11.6.2 Activity: Error Analysis

Component Overview: Students examine four pieces of student work to determine what each student did correctly and incorrectly. After explaining the correct and incorrect steps each student took, students solve each problem with their partner.



11.6.2 Activity: Error Analysis

With your partner, answer each question below.

1. Diana was given the following problem to solve.

Eugene has \$40.00 to spend at a theme park gift shop. He receives a discount of $\frac{1}{5}$ off the price of an item for being an annual passholder. He wants to purchase a collectible passholder pin that has a retail price of \$19.95. How much will he have left after purchasing the pin?

Diana solved the problem with the steps shown.

Diana's Work

$$\begin{array}{r}
 19.95 \cdot (-\frac{1}{5}) \\
 19.95 \cdot (-\frac{1}{5}) \\
 \hline
 -19.95 \\
 \hline
 5 \overline{)19.95} \\
 \underline{-15 } \\
 49 \\
 \underline{-45 } \\
 45 \\
 \underline{-45 } \\
 0
 \end{array}$$

$$\begin{array}{r}
 39.9 \\
 \underline{-3.99} \\
 36.01
 \end{array}$$

Eugene had \$36.01 left.

- a. Identify what Diana did correctly and incorrectly.

Correct	Incorrect
Diana determined that the discount from being an annual passholder was \$3.99.	Instead of taking the discount from the cost of the collectable passholder pin, Diana subtracted the discount from the total amount Eugene had to spend.

- b. How much money will Eugene have left over after buying the pin? Show your work.

\$24.04



MA.K12.MTR.4.1: Discussion and Reflection

Continue encouraging students to analyze the work of others throughout this activity. Consider what things you look for as a teacher and prompt students to utilize the same strategies for analysis.



Task Cards

Consider setting up the student work examples as Task Cards for students to rotate in groups or stations. It may be helpful to group students based on varying mastery levels to ensure groups do not get stuck and students can collaborate effectively to solve each problem.



In order to assist students with analyzing student work, urge them to number and write out the steps from each student example. This provides opportunities for students to use specific mathematical language to communicate correct and incorrect steps and calculations.



Think-Pair-Share

Consider engaging students in a Think-Pair-Share routine as they identify what each student did correctly and incorrectly for questions 1-4. Position students to analyze and build on each other's responses before putting their answers in writing.

11.6.2 Activity: Error Analysis (continued)

2. Jeffrey was given the following problem to solve.

Diego is planning to hike the very difficult Shorty's Well Route to Telescope Peak, which begins at an elevation of -262 feet and ends at an elevation of $11,043$ feet. He wants to hike $\frac{3}{5}$ of the change in elevation of the trail on the first day of his hike. If Diego sticks to his plan, at what elevation will he start on the second day?

Jeffrey solved the problem with the steps shown.

Jeffrey's Work

$$11043 - 262 =$$

$$\begin{array}{r} 11043 \\ - 262 \\ \hline 10781 \text{ feet} \end{array}$$

$$10781 \cdot \frac{3}{5} = \frac{3234}{5}$$

$$\begin{array}{r} 6468.6 \text{ feet} \\ 5 \overline{)32343.0} \\ \underline{-30} \\ 23 \\ \underline{-20} \\ 34 \\ \underline{-30} \\ 43 \\ \underline{-40} \\ 30 \\ \underline{-30} \\ 0 \end{array}$$

Diego plans to be at 6468.6 feet to start Day 2.

a. Identify what Jeffrey did correctly and incorrectly.

Correct	Incorrect
Jeffrey understood that he needed to multiply the change by $\frac{3}{5}$.	Jeffrey first needs to find out how much elevation the trail changes. He should have subtracted the highest elevation from the lowest to find this. His subtraction has an error as he should have added the opposite of -262 .

b. At what elevation will Diego start on the second day? Show your work.
6,521 feet



Students may struggle to represent Diego's starting elevation because it is below sea level and will therefore have trouble finding the error in Jeffrey's work. Ask questions to further scaffold this problem:

- What does it mean if the starting elevation is -262 feet?
- How can you represent a change in elevation when the starting point is below sea level and the end point is above sea level?

11.6.2 Activity: Error Analysis (continued)

3. Ashlyn was given the following problem to solve.

Fernando runs after school to prepare for the cross-country season. On Monday he ran 4.34 miles, Wednesday he ran 6.56 miles, and on Friday he ran 1.375 miles. His teammate Marcus only ran $\frac{4}{10}$ the number of miles as Fernando. How many miles did Marcus run?

Ashlyn solved the problem with the steps shown.

Ashlyn's Work	
$\begin{array}{r} 4.34 \\ +6.56 \\ \hline 10.90 \\ +1.375 \\ \hline 23.65 \text{ miles} \end{array}$	$\begin{array}{l} 23.65 \div \frac{4}{10} \\ 23.65 \cdot \frac{10}{4} = \frac{236.5}{4} \\ \begin{array}{r} 59.0 \\ 4 \overline{)236.5} \\ \underline{20} \\ 36 \\ \underline{36} \\ 0 \end{array} \end{array}$
	Fernando ran 59 miles.

- a. Identify what Ashlyn did correctly and incorrectly.

Correct	Incorrect
Ashlyn correctly added Fernando's miles from Monday and Wednesday.	Ashlyn made two mistakes. She incorrectly added 10.90 and 1.375. Also, she should have multiplied by $\frac{4}{10}$ instead of dividing. She also mislabeled her answer as how many miles Fernando ran when she was looking for how many miles Marcus ran.

- b. How many miles did Marcus run? Show your work.
4.91 miles



It may be helpful to provide a visual entry point for each problem where applicable. For example, question 3 describes the number of miles a student ran. Encourage students to represent this problem using a tape diagram to provide a concrete visual, or model this for students who are struggling to conceptualize the problem.

11.6.2 Activity: Error Analysis (continued)

4. Craig was given the following problem to solve.

Rhonda makes a specific hair dye color by mixing $1\frac{1}{8}$ ounces (oz) of red tone, 0.75 oz of gray tone, and $\frac{5}{8}$ oz of brown tone. If Rhonda needs to make 20 oz of the same color dye, how much of each color tone is needed?

Craig solved the problem with the steps shown.

Craig's Work

$$\begin{array}{r} 1\frac{1}{8} + \frac{5}{8} \\ \frac{8}{8} + \frac{5}{8} = \frac{13}{8} \\ \frac{13}{8} = \frac{5}{4} \end{array}$$

$$\begin{array}{r} \frac{5}{4} + 0.75 \\ 1\frac{1}{4} + 0.75 \\ 1.25 + 0.75 \\ + 0.75 \\ \hline 2.00 \text{ ounces total} \end{array}$$

$$20 \div 2 = 10 \text{ ounces needed of each}$$

$$\begin{array}{l} 1\frac{1}{8} \cdot 10 = 11\frac{2}{8} \text{ oz red} \\ 0.75 \cdot 10 = 7.5 \text{ oz gray} \\ \frac{5}{8} \cdot 10 = \frac{50}{8} = 6\frac{2}{8} \text{ oz brown} \end{array}$$


What advice would you give Craig to avoid making the same error in the future?

- a. Identify what Craig did correctly and incorrectly.

Correct	Incorrect
Craig understood that he needed to first find the number of ounces in 1 batch of dye before finding out how much of each they need for 20 oz.	Craig made an error in rewriting the mixed number as a fraction before adding in the first step. Craig also made an error simplifying $\frac{13}{8}$.

- b. How much of each color is needed to make 20 oz of dye? Show your work.

Red Tone - $1\frac{1}{8} \cdot 8 = 9 \text{ ounces}$

Gray Tone - $0.75 \cdot 8 = 6 \text{ ounces}$

Brown Tone - $\frac{5}{8} \cdot 8 = 5 \text{ ounces}$

11.6.3 Collaboration: Finding Missing Values

Component Overview: Students analyze a utility bill with missing information and work with a partner to determine the missing information. Students then work with a new partner to compare their missing values and resolve any differences.



11.6.3 Collaboration: Finding Missing Values

Many homeowners have solar panels that can generate energy to power their homes. The Better Utility Company charges customers who have solar panels using the following criteria:

- Customers are charged \$0.12 per kilowatt-hour for the energy used that is provided by the utility company.
- The utility company does not charge customers for energy they use from their own solar panel(s).
- For every kilowatt-hour of electricity that customers with solar panels generate but do not use, the utility company gives a price adjustment of $-\$0.025$.

Erin Hill's bill for the month of July is missing some information.

Best Utility Company		Erin Hill	
123 Ohms Way		456 Long St.	
Big City, ST 12345		Small Town, ST 98765	
Kilowatt-hours used from Best Utility Company	683.67	Charge	\$82.04
Kilowatt-hours generated by solar panel and not used by customer	164	Credit	$-\$4.10$
		Amount Due	\$ 77.94

1. Work with your partner to fill in the missing information on the bill. Round answers to nearest hundredth.
2. Find another classmate other than your partner and compare your missing values.
 - If the missing values for both are the same, have your classmate initial next to each value in your bill.
 - If the missing values are different, discuss your work to determine what error(s) were made. After correcting any mistakes, initial next to each value in each other's bill.



Consider explaining to students that energy usage is measured in kilowatt-hours and that \$0.12 per kilowatt-hour is the cost for each unit for this utility company. Kilowatt-hour doesn't mean the number of kilowatts being used per hour. It is the power consumption of 1000 watts sustained for one hour. Using a laptop all day is approximately 1 kilowatt-hour of energy. Kilowatt-hour is a measure of the energy consumed.



Present the kilowatt-hours used over 4 weeks and the kilowatt-hours generated over 4 weeks. Allow the students to calculate the total kilowatt-hours of each over the four weeks. By incorporating an additional operation of addition with the division and subtraction operations already present, students will strengthen the connection to the objective of using three operations.



Prior to completing question 2, assist students with determining errors and correcting mistakes with their classmates. Provide questions that students can ask each other when comparing (and contrasting) values, such as:

- What values and operation did you use to find the kilowatt-hours?
- What values and operation did you use to find the amount due?
- What steps did you follow to reach your answer?

11.6.4 Wrap-Up: Reflection

Component Overview: Students will complete a self-reflection about the day's lesson.



11.6.4 Wrap-Up: Reflection

- Complete the table by reflecting on the day's lesson.

Answers will vary.

Today I deserve  for ...	
Today I  by doing ...	
Today I was  by ...	



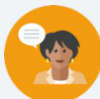
Post information about what each image is representing close to each image. This will assist students with what responses are expected —the “hand clapping” image is to elicit a positive aspect about the lesson and the “road signs” image is to elicit a confusing aspect about the lesson.

Alternatively, students can edit the image provided or sketch their own image and include their rationale for why they chose that image.

11.7 Solving Real-World Problems With Rational Numbers – Part 3

Lesson Introduction

 Benchmark	MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers.		 Learning Target	I can solve problems with at least three operations involving any of the four operations with rational numbers.
 Benchmark Coherence	Building On		Working Toward	
	MA.6.NSO.2.3 Solve multi-step, real-world problems involving any of the four operations with positive multi-digit decimals or positive fractions, including mixed numbers.		MA.8.NSO.1.7 Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers including exponents and radicals.	
 Essential Questions	How do you solve multi-step real-world problems involving any of the four operations with rational numbers?			
 Lesson Narrative	This lesson applies skills students have used throughout the unit to solve various real-world scenarios using operations with rational numbers. Students work in small groups to analyze and answer questions about a given scenario, then present their work to the rest of the class through a gallery walk. This lesson culminates in a self-reflection on student understanding of using operations with rational numbers.			
 Component Summary	11.7.1 Warm-Up (~5 min.)	Analysis of image of office supplies (<i>SE p. 196</i>)	11.7.3 Your Turn! (10min.)	Practice performing operations with rational numbers (<i>SE p. 198</i>)
	11.7.2 Activity (25 min.)	Gallery Walk – exploring real-world contexts for problem-solving (<i>SE p. 196</i>)	11.7.4 Wrap-Up (~5 min.)	Self-reflection on level of understanding (<i>SE p. 199</i>)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - N/A <u>Additional Terms:</u> - Balance		- Chart paper - Calculator (Optional) Ruler	
			Additional Preparation	
			Gallery Walk – provide each scenario on chart paper for students to rotate while exploring. For smaller group sizes, make 2 copies of each scenario, splitting the class into “A” and “B” rotations in which each letter contains all four scenarios.	

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may use the incorrect operation(s) to reach the answer - Students may misunderstand vocabulary used in the real-life scenarios.	Consider how students can add creative elements to the gallery walk. Encourage students to utilize diagrams, tables, pictures, or other creative elements. Students can use construction paper to showcase different elements.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
 Differentiate Instruction	Notice and Wonder The 11.7.1 Warm-Up presents a “Notice and Wonder” routine. Students will examine the image and will state what they notice and what they wonder about from the image presented.	MA.K12.MTR.7.1: Real-World Contexts This standard emphasizes experiencing math through real-world examples. Students will be exploring various real-world examples that use various forms of rational numbers and will perform all operations. Provide opportunities for students to make connections between each context and their personal experiences.
	Consider providing translations of each context for English Language Learners, or adjust with cultural intelligence in mind, affording students in the early stages of language acquisition development the opportunity to create their own contexts and problems based on their experiences and cultural capital.	
	Additionally, offer all students a variety of reading comprehension strategies to employ during the Gallery Walk, making use of tools and resources for analyzing and unpacking each context. - Three Reads: Instructional routine for targeting specific information - Mark-Up Text: Visual strategy for identifying important information - Partner Read: Auditory scaffolding for students to engage verbally in supporting different entry points	
Lesson Notes:		

11.7.1 Warm-Up: Notice and Wonder

Component Overview: An image of binder clips and a stack of papers is given. Students will analyze the image through a “Notice and Wonder” routine and will answer questions regarding the size of the binder clips in reference to sheets of papers.



11.7.1 Warm-Up: Notice and Wonder

1. What do you notice about the binder clips in the image?

Answers will vary. There are three sizes of binder clips pictured – small, medium, and large. The binder clips are in order from right to left and the largest clip is on the right and they are going down in size to the left.



2. What questions come to mind when you look at this image?

Answers will vary. Which binder clip can hold the stack of paper pictured? What is the maximum number of pages that each binder clip can hold?

3. The stack of papers contains about 260 sheets of paper. Estimate the number of sheets each binder clip can hold.

Answers will vary.

<i>Small Binder Clip</i>	<i>Medium Binder Clip</i>	<i>Large Binder Clip</i>
<i>75</i>	<i>175</i>	<i>275</i>

4. To make a good estimate, what information would be useful to know?

Answers will vary. How many inches does each binder clip hold? How many inches is a sheet of paper?

5. What tools or resources could help you find that information?

Answers will vary. One tool that would be helpful with finding the information is a ruler to measure the width of the binder clip opened.



Notice and Wonder

The Warm-Up presents an opportunity for students to use mathematical language in order to communicate suitability of different rational numbers in real-life situations. Engage the class in a Notice and Wonder routine to foster discussion. Students may notice that the width of each paper clip could be described using decimals. Students might comment that fractions of a ream of paper could be used to communicate how many sheets of paper each binder clip can hold. Students might also state that negative rational numbers are not applicable in this situation.

11.7.2 Activity: Gallery Walk

Component Overview: In this Gallery Walk, students work with group members to answer questions about presented scenarios. Display each scenario on chart paper throughout the room and assign groups which scenario to complete. Once students have answered the questions assigned, students will present their work to the rest of the groups.

11.7.2 Activity: Gallery Walk

Your teacher will assign your group one of the following scenarios. Work together to answer the question(s), writing neatly on the chart paper provided. Show all your work to prepare a presentation for the other students in your class.

Scenario 1	Marie plans to use a ride sharing service for the trip from Tampa International Airport to her hotel. The service has a booking fee of \$2.45 and a cost of \$0.42 for the first $\frac{1}{4}$ mile and an additional fee of \$0.16 for each 0.1 mile after the first $\frac{1}{4}$ mile.
	1. How much will the trip cost if Marie's hotel is 7.25 miles from the airport?
	$\$2.45 + \$0.42 + \left[(\$0.16) \left(\frac{7}{1} \right) \right] = \14.07
	2. Marie decides to stop by a grocery store before arriving at her hotel. The ride sharing service charges \$0.45 for each minute they have to wait on a customer. Her total bill was \$21.72. How long was Marie in the grocery store?

$$21.72 - 14.07 = 7.65$$

$$7.65 \div 0.45 = 17 \text{ minutes}$$



MA.K12.MTR.7.1: Real-World Contexts
Model connections to personal experiences for students to actively engage with applications of mathematics. Consider having students answer these questions on their chart paper:

- Is this real-world context something you have experienced?
- What experiences have you had that are similar to this real-world context?



Gallery Walk

Have small groups of students work one of the assigned questions on chart paper. When all groups are done, post the chart paper around the room. Have students "Gallery Walk" to each question, adding comments or suggested error corrections to the work. If the class is large, make 2 copies of each question for smaller working groups.



What form of rational numbers will each answer be?

11.7.2 Activity: Gallery Walk (continued)

Scenario 2

Leah and her parents have an agreement. She borrowed money from them to buy a video gaming system and is supposed to pay them back within 3 months. If Leah does not have the balance paid by the agreed-upon time, she will have to give the system to her parents until it is paid off. Leah and her parents created a payment plan to help her meet this goal. Starting on July 1, Leah should pay \$5.95 per day until the gaming system is paid off. On July 5th, her balance was -\$10.25 since she couldn't pay the full amount each day. Leah was not able to pay anything again for 15 days. On July 20th, Leah paid \$85.00.

1. How much had Leah paid her parents by July 5th?

$$(\$5.95)(5) + (-\$10.25) = \$19.50$$

2. By July 20th, how much more or less had Leah paid than what was required by the payment plan?

$$(\$5.95)(20) + (-\$19.50) + (-\$85) = \$14.50$$

Leah had to pay \$14.50 more on her payment plan.



Students may struggle with whether to include 4 or 5 days in question 1. Ask students which they consider is best and why. Modify your key based on student choice. Inform these students that they should be consistent with their choice as they answer question 2.



Consider capturing the cultural intelligence or cultural capital of your students that are unique to their class and their peers and applying that capital for teaching and learning.

Is there a local coffee shop or tienda to use as an anchor or reference? Consider providing a picture or actual worksheet of their changes to better engage students.

11.7.2 Activity: Gallery Walk (continued)

Scenario 3

A coffee shop keeps track of changes in their coffee inventory. A week of their record-keeping is shown.

Day	Change in Inventory (lb)
Monday	150
Tuesday	−43.3
Wednesday	−50.5
Thursday	72.8
Friday	−52.7
Saturday	−21.6
Sunday	0

Coffee is delivered in 50-pound (lb) bags, 3 bags at a time.

- How much coffee is left at the end of the week?

54.7 lbs

- Coffee is delivered in 50-pound (lb) bags, 3 bags at a time. How many pounds of coffee were sold on Thursday?

$150 - 72.8 = 77.2$ pounds



Extend student thinking to interpret the changes by day. Based on the data in this table:

- Which day might the shop be most busy?*
- Is there a day the coffee shop may be closed? How do you know?*
- Should the shop order more or less on certain days of the week?*



Consider challenging students to consider sociocultural contexts that could facilitate meaningful discussions.

- Should energy be a free resource provided by the government? Why or why not?*
- Should solar energy be required by all homes? Why or why not? What challenges would there be?*
- If solar energy costs more money but conserves the environment, is it worth it? Why or why not?*

11.7.2 Activity: Gallery Walk (continued)

Scenario 4

Recall from the previous lesson that The Better Utility Company charges \$0.12 per kilowatt-hour for the energy a customer uses that is provided by the utility company. The company gives an adjustment of $-\$0.025$ for every kilowatt-hour of electricity a customer with solar panel(s) generates that they do not use themselves.

The table shows the number of kilowatt-hours of the electricity a customer used and generated each week for a month.

	Kilowatt-Hours Used	Kilowatt-Hours Generated that were not Used by the Customer
Week 1	112.08	253.2
Week 2	181.5	357.6
Week 3	101	308
Week 4	200.42	214.4

- What is the amount due for this month?
 $(595)(\$0.12) + (1133.2)(-\$0.025) = \$43.07$

	Used	Not Used	Total
Week 1	112.08	253.2	\$7.12
Week 2	181.5	357.6	\$12.84
Week 3	101	308	\$4.42
Week 4	200.42	214.4	\$18.69
- The average customer's bill is $\frac{2}{3}$ of what they would have been charged if they had not produced electricity using solar panels. Explain if this customer's savings was below or above that average.
Answers will vary. This customer paid about 60.3% of what they would have paid, which is less than $\frac{2}{3}$ of the original amount, so they saved more than the average customer who has solar panels.

11.7.3 Your Turn!

Component Overview: Students will answer two questions about real-life situations involving operations with rational numbers.



11.7.3 Your Turn!

- Araya is making red ribbons to sell to support a local nonprofit organization. She purchased a roll of fabric that is $30\frac{3}{4}$ feet in length and each ribbon uses 0.635 feet of fabric.

If she already has 16 ribbon orders, how many additional ribbons will she be able to sell with the single roll? Show your work.

$$\left(30\frac{3}{4} \div 0.635\right) - 16 = 32.425 \quad \text{32 Ribbons}$$

- Joanne purchased $2\frac{1}{4}$ pounds of tomatoes, $3\frac{1}{8}$ pounds of red onions, $\frac{3}{4}$ pounds of asparagus, and 3 lemons from a grocery store.

Produce	Price
Tomatoes	\$2.99 per pound
Red Onions	\$1.79 per pound
Asparagus	\$2.49 per pound
Lemons	\$0.79 each

Based on the listed prices, how much would Joanne spend?

Approximately \$16.56



Offer students the option to perform operations using one form of rational numbers. Allow students to convert all mixed numbers into decimals and complete, and also allow students to convert all decimals into fractions or mixed numbers and complete.



In order to assist students with organizing calculations throughout problem solving for question 2, encourage students to add an additional column on the table given. This column would be an organized visual to record the total price of each item purchased.

11.7.4 Wrap-Up: Reflection

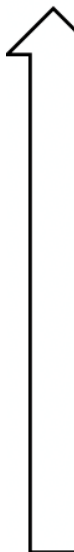
Component Overview: A self-reflection is presented for students to reflect on their level of current understanding of the lesson.



11.7.4 Wrap-Up: Reflection

1. Color in the arrow up to the statement which best describes your current understanding of solving real-world problems with rational numbers.

Answers will vary.



I'm so confident, I could explain this to someone else!

I can get to the right answer, but I don't understand well enough to explain it yet.

I understand some of this, but I don't understand all of it yet.

I tried hard and I listened, but I am finding this challenging. I will make sure that I get help.

I do not understand any of this yet. There are things I could do to be a better learner.



Consider having students provide examples of their level of understanding using a problem from the lesson. Students who are confident can write an explanation of how they solved. If they're struggling, students can circle or name which part of the problem they have questions about.

Note for growth mindset – avoid students naming things “they are not good at” or “they can’t do.” Language is important; therefore, students instead should use language like “Things I have questions about” or “I need help with.”

11.8 Evaluating Numeric Expressions – Part 1

Lesson Introduction

 Benchmarks	MA.7.NSO.2.1 Solve mathematical problems using multi-step order of operations with rational numbers including grouping symbols, whole-number exponents, and absolute value. MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.			 Learning Target	I can evaluate numeric expressions with up to four operations.
 Benchmark Coherence	Building On MA.6.AR.1.3 Evaluate algebraic expressions using substitution and order of operations. MA.6.NSO.1.4 Solve mathematical and real-world problems involving absolute value, including the comparison of absolute value.			Working Toward MA.8.NSO.1.7 Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers including exponents and radicals.	
 Essential Questions	- How can I use order of operations to evaluate numeric expression? - How are numeric expressions with multiple grouping symbols evaluated?				
 Lesson Narrative	In this lesson, students extend their knowledge of the order of operations to include what they have learned about exponents, the use of the absolute value symbol, and multiple grouping symbols. Students begin by getting a refresher on the process of the order of operations that they learned in 5 th grade and receive formal instruction on using the order of operations with exponents and grouping with a focus on absolute value notation. Students then get the opportunity to practice applying what they have learned by evaluating numeric expressions.				
 Component Summary	11.8.1 Warm-Up (~5 min)	Math symbols activity (<i>SE p. 200</i>)	11.8.4 Exploration (10-15 min)	Collaborative exploration activity on grouping symbols (<i>SE p. 203</i>)	
	11.8.2 Guided Instruction (10-15 min)	Instruction on the order of operations with exponents, absolute value, and grouping symbols (<i>SE p. 200</i>)	11.8.5 Your Turn! (5-10 min)	Independent practice on evaluating expressions (<i>SE p. 204</i>)	
	11.8.3 Collaboration (10 min)	Collaborative practice with evaluating expressions (<i>SE p. 203</i>)	11.8.6 Wrap-Up (~5 min)	Students reflect on their learning (<i>SE p. 205</i>)	
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> - Nested grouping symbols - Absolute value notation <u>Additional Terms:</u> - Order of operations		(Optional) Colored markers or highlighters (Optional) Calculators (Optional) Video		- Consider grouping for 11.8.3 Collaboration. - The 11.8.1 Warm-Up video is optional.

 Teacher Tips	Common Misconceptions - Students may confuse the order of operations by doing all multiplication before division instead of from left to right in order. - Students may forget to complete the calculation inside the absolute value before writing its actual absolute value. - Students may forget to write the positive value when working with absolute values. - Students may misunderstand the hierarchy of operations.	Things to Consider - Students may have trouble with nested grouping symbols and identifying which numbers/operations are in the inner grouping vs the outer grouping. Consider using colored highlighters or other visual markers to help students see the process unfold. - Often, there is more than one way to evaluate a numeric expression by applying the order of operations, properties of equality, and laws of exponents. As you work the problems, consider alternate correct ways to evaluate each expression to be prepared for alternate correct suggestions from students. - The order of operations is often taught using a mnemonic device as a hard and fast rule. The lesson does not contain any mnemonic devices but instead relies upon both the order of operations and the properties of equality, such as commutative and associative properties. Encourage students' flexible thinking by demonstrating various ways a numeric expression can be evaluated. - If time is an issue, then consider assigning 11.8.5 Your Turn! for homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Compare and Connect In 11.8.2 Guided Instruction, students learn about evaluating expressions when nested grouping symbols are present. To help students understand this, consider demonstrating thinking out loud and prompting students to reflect and respond.	MA.K12.MTR.4.1: Discussion and Reflection During 11.8.2 Guided Instruction, students have the opportunity to analyze the mathematical thinking of two students. Discussing the two different methods with their partner helps students construct a justification that reflects the two correct methods. In 11.8.4 Exploration, students work with their partner to explore similar numeric expressions where they communicate the order of operations, properties of equality, and laws of exponents.
	11.8.4 Exploration have students working together in partner pairs. Consider pairing students based on the data from the previous lesson. This can help struggling students learn from their peers while strengthening the other students' understanding of the concept. 11.8.6 Wrap-Up activity in this lesson can provide valuable data on whether or not a student understands the concepts in this lesson.	
Lesson Notes:		

11.8.1 Warm-Up: Math Symbols

Component Overview: Students will engage in an activity on math symbols where they list symbols they are familiar with. They will distinguish between ones they are familiar with and symbols they have seen but are unfamiliar with.



11.8.1 Warm-Up: Math Symbols

In the 18th century, Robert Recorde decided to use a symbol for the idea of “is equal to” so he would not have to write the words over and over. As more mathematicians also began to use the symbol, the “=” became accepted as the standard symbol for equality. Math is full of symbols that shorten lengthy words and calculations. But just like symbols you use when communicating with others, like “LOL” in texting, once you learn their meaning, understanding the mathematics is not so challenging.

1. List math symbols that you are familiar with and what they represent.

Answers will vary. $\times$ means multiply, $-$ means subtract, $+$ means add.

2. List other math symbols you have seen before, but you are unfamiliar with.

Answers will vary. $\pm$, $\cup$, $\approx$



Consider providing students with several symbols to choose from. Students may not be able to recall symbols that they have seen but do not understand.

11.8.2 Guided Instruction: Applying the Order of Operations for Expressions Including Exponents

Component Overview: Students will receive formal instruction on using the order of operations with exponents, absolute value, and nested grouping symbols. For this activity, consider leveraging the “I Do” and “We Do” components of the gradual release model. Facilitate the questions in a discussion form to help students connect to the concepts.



11.8.2 Guided Instruction: Applying the Order of Operations for Expressions Including Exponents

Recall from 5th grade that an order of operations is used to evaluate numeric expressions with multiple operations.

Step 1: Perform operations within grouping symbols.

Step 2: Multiply and divide in order from left to right ($\rightarrow$).

Step 3: Add and subtract in order from left to right ($\rightarrow$).

Numerical expressions can also include exponents.

1. The expression $4 \times (-2)^3 - 8 \div (-3 + 1)$ can be rewritten by expanding the exponential expression as $4 \times (-2 \cdot -2 \cdot -2) - 8 \div (-3 + 1)$.
 - a. According to the order of operations, which expressions should be evaluated first?
The expressions in parentheses should be evaluated first.
 - b. When $(-2)^3$ is rewritten using multiplication, the resulting expression is in parentheses. Complete the statement.

According to the order of operations, $(-2)^3$ should

be evaluated ☐ after multiplying by 4.

☒ before



Students will spend more time evaluating expressions with exponents in Unit 13 Algebraic Expressions. The goal here is to make a connection to exponents as grouped repeated multiplication to help students understand why exponents should be evaluated before multiplication and division in a numeric expression.

11.8.2 Guided Instruction: Applying the Order of Operations for Expressions Including Exponents (continued)

- c. Work with your partner to number each operation so that the order of operations is in the correct order.

- 1 Perform operations inside grouping
- 3 Perform multiplication and division in order from left to right. ($\rightarrow$)
- 2 Evaluate expressions with exponents.
- 4 Perform addition and subtraction in order from left to right. ($\rightarrow$)

2. Use order of operations to evaluate one of the expressions shown in the table below while your partner evaluates the other expression.

Expression A	Expression B
$(6 + 0.5)^2 \div 0.25$	$(6^2 + 0.5^2) \div 0.25$
$6.5^2 \div 0.25$ $= 42.25 \div 0.25$ $= 169$	$(36 + .25) \div 0.25$ $= 36.25 \div 0.25$ $= 145$

3. How are the problems similar?

Answers will vary. The problems are similar by both having exponents and the final step being to divide by 0.25.

4. How are the problems different?

Answers will vary. The problems are different in that in expression A, you add the expressions inside the parenthesis before applying the exponent, whereas in expression B you apply the exponents to the expressions inside the parentheses before taking their sum.



For questions 2 and 3, consider having students discuss them with a partner and then share out their responses in whole group. The responses may be more thorough when provided orally from students than their writing of the explanation in the book.

11.8.2 Guided Instruction: Applying the Order of Operations for Expressions Including Exponents (continued)

5. Complete the statements about the expressions.

The two expressions are ☐ equivalent
☒ not equivalent

because they ☐ have the same value.
☒ do not have the same value.

For Expression A, first

- ☒ add inside the parentheses, then apply the exponent to the sum.
- ☐ apply the exponent to each addend, then add to find the sum.

For Expression B, first

- ☐ add inside the parentheses, then apply the exponent to the sum.
- ☒ apply the exponent to each addend, then add to find the sum.

For both expressions, addition was done before division

because ☐ the addition is to the left of the division.
☒ the addition is inside the grouping symbols.

6. Anna evaluated Expression A and Expression B. Her work is shown.

Expression A	Expression B
$ -10 \div 2 + 10^2$	$-10 \div 2 + 10^2$
$10 \div 2 + 10^2$	$-10 \div 2 + 100$
$10 \div 2 + 100$	$-5 + 100$
$5 + 100$	95
105	



Students may need additional strategies for how to approach these types of questions. Consider showing students how to break down each sentence in this series of statements and how to look for correct matches in each one.



How can you determine if two numerical expressions are equivalent?



- *What would have happened if the 10 was positive in both?*
- *Would the answers have been different still?*
- *Why or why not?*

11.8.2 Guided Instruction: Applying the Order of Operations for Expressions Including Exponents (continued)

Anna's partner gave her a thumbs up on both of her answers. Why did Anna arrive at two different values when evaluating these similar looking expressions?

In expression A, you have to find the absolute value of -10, making the value a positive 10.



Absolute value notation should be treated the same as grouping symbols when following the order of operations.



Consider providing additional time to discuss the absolute value component. Some students may have forgotten how absolute value works and need a minor refresher when reviewing.

7. Zion says since multiplication is commutative and the order does not matter, he thinks he can evaluate the expression $[0.05 \times (3 \times 2)^4]^2$ by either multiplying 0.05×3 first or by multiplying 3×2 first.

- a. Discuss with your partner if Zion is correct when he stated that both would work.

Answers will vary. The result of the discussion should be that Zion is incorrect. There is an exponent that has to be applied first in this case, so you must multiply 3 by 2 and then raise the result to the 4th power before multiplying by 0.05. Multiplication is commutative; however, what Zion is describing is actually the associative property.

- b. Evaluate the expression $[0.05 \times (3 \times 2)^4]^2$.

$[0.05 \times 6^4]^2$
 $[0.05 \times 1,296]^2$
 $[64.8]^2$
 4,199.04



When using the order of operations to evaluate an expression with nested grouping symbols, apply the order of operations inside the inner grouping symbols first.



Some students may struggle when working with nested grouping symbols. To help with this, have students mark the outer grouping symbols with green highlighter and underline everything between them. Then have them mark the inner grouping symbols with a yellow highlighter and draw a line above the symbols.

11.8.3 Collaboration: Determining Equivalent Expressions

Component Overview: Students will work collaboratively with evaluating expressions. For this activity, students should work with a partner or small group to complete.



11.8.3 Collaboration: Determining Equivalent Expressions

Answer the questions with your partner.

Tanisha says that when she evaluated an expression, the result was 7.25.

1. Use mental math to estimate each value. Then, circle which of the following expressions could have been Tanisha's expression.

$$(6 - 0.5)^2 \times 5$$

$$6 - 0.5^2 \times 5$$

$$6 + (-0.5)^2 \times 5$$

$$5 \times 6 - 0.5^2$$

2. Find the values of the expressions to verify your choice.

Expression	Value
$(6 - 0.5)^2 \times 5$	151.25
$6 - 0.5^2 \times 5$	4.75
$6 + (-0.5)^2 \times 5$	7.25
$5 \times 6 - 0.5^2$	29.75



Consider asking students which expressions in question 1 are equivalent. This may give some insight on how much the student has already learned.



Consider having students turn their calculators over for question 1 and then to use their calculators for question 2. Calculators such as Desmos can determine the value if the student inputs the expression as written. Consider providing students with four-function calculators, where the student will have to use their knowledge to evaluate while the calculator can be a support to learning.

11.8.4 Exploration: Grouping Symbols

Component Overview: Students will work collaboratively to evaluate expressions involving grouping symbols. This activity is designed for students to complete with a partner.



11.8.4 Exploration: Grouping Symbols

- Without working the problems, discuss with your partner which expression in the table you expect to have the largest value.

Answer will vary. Discussion should be around law of exponents and order of operations to result in the largest value.

Expression A	Expression B
$10 - (0.75 \times 12) + 4^2$ $10 - 9 + 4^2$ $10 - 9 + 16$ 17	$(10 - 0.75) \times (12 + 4^2)$ $(9.25) \times (12 + 16)$ 9.25×28 259
Expression C	Expression D
$10 - 0.75 \times 12 + 4^2$ $10 - 0.75 \times 12 + 16$ $10 - 9 + 16$ 17	$10 - 0.75 \times (12 + 4)^2$ $10 - 0.75 \times (16)^2$ $10 - 0.75 \times 256$ $10 - 192$ -182

- Put an asterisk (*) next to the expression you expect will have the largest value. Write your explanation here.

Answer will vary. Explanation will be based on the laws of exponents or order of operations used.

- Evaluate each of the expressions in the table with your partner.



Prior to releasing this activity to students, consider reviewing the narrative with the students. Many students will jump straight to the expressions and begin evaluating and miss the key point of them making a guess as to which one they think will have the largest value. You may also want to consider having the class vote and writing the total number of votes per expression in a central area to refer back to when reviewing the answers.

11.8.4 Exploration: Grouping Symbols (continued)

4. With another set of partners, share if your guess was or was not correct. Discuss how the order of operations affected the value of your guesses. Summarize your discussion.

Answers will vary. The discussion should be about order of operations causing expression D to be the largest value.

5. For the two expressions in the table that resulted in the same value, explain why they have the same value.

Where the grouping symbol was placed did not have any effect on the order of operations. Multiplication would have been completed second in both expressions after evaluating the exponent.



What was the impact of the parentheses on the four expressions in this activity?

11.8.5 Your Turn!

Component Overview: Students will practice evaluating expressions with multiple steps using the order of operations. For this activity, students should work independently to practice what they've learned in this lesson.



11.8.5 Your Turn!

1. Evaluate each expression without using a calculator.

a. $[(10 - 15) \times 2]^3 + 50$

$$[-5 \times 2]^3 + 50$$

$$(-10)^3 + 50$$

$$-1000 + 50$$

$$-950$$

b. $(2^4 - 3 \times 6)^2$

$$(16 - 3 \times 6)^2$$

$$(16 - 18)^2$$

$$(-2)^2$$

$$4$$

c. $|18 + (-2) \times 14|^5$

$$|18 + (-28)|^5$$

$$|-10|^5$$

$$10^5$$

$$100,000$$

d. $[0.8 \times (1.2 + 2.8)^2]^2$

$$[0.8 \times 4^2]^2$$

$$[0.8 \times 16]^2$$

$$12.8^2$$

$$163.84$$

e. $[(0.8 \times 10)^2 + 4.5] \div 0.5$

$$[8^2 + 4.5] \div 0.5$$

$$[64 + 4.5] \div 0.5$$

$$68.5 \div 0.5$$

$$137$$

f. $(10 \times 0.4)^2 + (28.4 \div (-4))$

$$4^2 + (28.4 \div (-4))$$

$$4^2 + (-7.1)$$

$$16 + (-7.1)$$

$$8.9$$



Watch for how students perform on question 1a during this activity. If they are performing poorly, consider pulling the students into small groups to work through one of the other questions and then releasing them to do the rest on their own.

11.8.6 Wrap-Up: Reflection

Component Overview: This activity should be done independently to provide specific student-level data.



11.8.6 Wrap-Up: Reflection

1. Complete each statement based on your learning today.

- A barrier to my learning was ...
Answers will vary.
- I will try to overcome this by ...
Answers will vary.
- Today I am still unsure about ...
Answers will vary.
- To fill in this gap I intend to ...
Answers will vary.

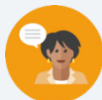


*If students are able to get the correct answer but not identify the mistake, they may not fully comprehend what is occurring incorrectly in someone else's work.
If students are not able to locate a mistake in the problem, it will highlight a specific misconception that they have.
If students identify the mistakes correctly but the response is not correct, they may have made a calculation error.*

11.9 Evaluating Numeric Expressions – Part 2

Lesson Introduction

 Benchmarks	MA.7.NSO.2.1 Solve mathematical problems using multi-step order of operations with rational numbers including grouping symbols, whole numbers including grouping symbols, whole-number exponents, and absolute value. MA.7.NSO.2.2 Add, Subtract, multiply, and divide rational numbers with procedural fluency.			 Learning Target	I can evaluate numeric expressions with up to six operations.
 Benchmark Coherence	Building On MA.6.AR.1.3 Evaluate algebraic expressions using substitution and order of operations. MA.6.NSO.1.4 Solve mathematical and real-world problems involving absolute value, including the comparison of absolute value.			Working Toward MA.8.NSO.1.7 Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers including exponents and radicals.	
 Essential Questions	- How can I evaluate numeric expressions with up to six operations? - How can I apply what I've learned in this unit to evaluate expressions with up to six operations?				
 Lesson Narrative	Lesson 9 builds on the previous lesson by providing more practice and by incorporating more steps to evaluate each expression. Students will be asked to evaluate expressions that contain several operations and will require them to apply all that they have learned throughout this unit to complete the evaluations.				
 Component Summary	11.9.1 Warm-Up (~5 min)	Bell Work evaluating expressions with grouping symbols (<i>SE p. 206</i>)	11.9.4 Try It (10 min)	Using the order of operations activity (<i>SE p. 208</i>)	
	11.9.2 Guided Instruction (5-10 min)	Evaluating expressions with six operations (<i>SE p. 206</i>)	11.9.5 Activity (10-15 min)	Class math debate activity (<i>SE p. 209</i>)	
	11.9.3 Activity (10-15 min)	Dice game with evaluating expressions (<i>SE p. 207</i>)	11.9.6 Wrap-Up (~5 min)	Self-reflection on today's lesson (<i>SE p. 209</i>)	
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> Order of operations		- Dice (alternate option is a spinner) (Optional) Paper to create labels for corners A, B, C, and D		11.9.5 Activity requires the corners of the room to be labeled A, B, C, or D. (Optional) For 11.9.4 Try It, take the table and cut it into strips with the step numbers, the expressions, and a premade explanation.

 Teacher Tips	Common Misconceptions - Students may misuse the order of operations by considering it a rule, not a guidance, that works with laws of exponents and properties of equality. - Students may make computation errors when evaluating. - Students may struggle with nested grouping symbols when there are several steps.	Things to Consider - Consider preparing your groupings and having them sit in those seats prior to launching the lesson in order to reduce movement. - Students should not use a calculator for this lesson. - If time is an issue, then consider 11.9.4 Try It for homework.
	Literacy and Language Routines Collect and Display 11.9.3 Activity has students roll dice to place values into a prewritten expression. Consider scribing what you see and hear to display for all students to view and use. This provides feedback for students in a way that increases accessibility while simultaneously supporting meta-awareness of language.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.3.1: Mathematical Fluency One of the primary goals of this lesson is for students to be able to apply what they've learned to evaluate numeric expressions. As such, students should be able to evaluate them with a high degree of fluency, since they have been working on this for two lessons. While the expressions may become lengthier, the process has remained the same.
	 Differentiate Instruction	
11.9.3 and 11.9.5 Activities have students working together with a partner or in small groups. Consider grouping students in a high/low performance grouping based on the data from the previous lesson. This can help struggling students learn from their peers while strengthening the other students' understanding of the concept. 11.9.5 Activity incorporates a great deal of movement for students. Additionally, it provides an opportunity for students to verbalize their thinking around choosing the correct expression. 11.9.4 Try It allows students to try using the order of operations in an independent setting. It also has them identify and explain the steps. Student performance on question 1 can highlight any potential misconceptions that may need to be addressed with an individual student.		
Lesson Notes:		

11.9.1 Warm-Up: Bell Work

Component Overview: Students will complete a Bell Work activity in order to explain the steps needed to evaluate the provided expressions. This activity is designed to be done independently.



11.9.1 Warm-Up: Bell Work

Explain the steps needed to evaluate the expressions.

1. $(10 + 18 \cdot 2)^5 \div 3$

Inside the parenthesis multiply 18 by 2, then add the product to 10. Take the value of that to the fifth power. Then, divide the result by 3.

2. $[(7 - 10)^2]^4 \times 2$

Inside the parenthesis, subtract 10 from 7. Then, take the value of that to the second power. Then, take the result to the fourth power. Lastly, multiply the value by 2.



With this activity, notice that these questions are not asking for students to evaluate them but rather explain the steps needed to evaluate.

In question 2, students could also state that the difference of $7 - 10$ is taken to the power of 8.



Consider creating this as a partner discussion activity where they can describe question 1 with their partner and then switch for question 2. Engage students in a discussion of the results by having them share out their steps either with one student per problem or having each student share the next step in evaluating the expression.

11.9.2 Guided Instruction: Evaluating Expressions with Multiple Steps

Component Overview: Students will receive formal instruction on evaluating expressions when there are six operations involved by exploring steps used by two different students.



11.9.2 Guided Instruction: Evaluating Expressions with Multiple Steps

Two students evaluated the expression below in two different ways. Review the work of each student. Then, answer the questions.

Diego	Valeria
$[0.5 \times (0.5 + 3.5)^2]^3 + 26 \div 13$	$[0.5 \times (0.5 + 3.5)^2]^3 + 26 \div 13$
$[0.5 \times 4^2]^3 + 26 \div 13$	$0.5^3 \times (0.5 + 3.5)^{2 \times 3} + 26 \div 13$
$[0.5 \times 16]^3 + 26 \div 13$	$0.5^3 \times (0.5 + 3.5)^6 + 26 \div 13$
$8^3 + 26 \div 13$	$0.5^3 \times 4^6 + 26 \div 13$
$512 + 26 \div 13$	$0.125 \times 4^6 + 26 \div 13$
$512 + 2$	$0.125 \times 4,096 + 26 \div 13$
514	$512 + 26 \div 13$
	$512 + 2$
	514

- How were Diego's steps and Valeria's steps different from each other?

Answers will vary. Diego used order of operations to simplify the expression whereas Valeria used the laws of exponents.

- Discuss with your partner whether you think one method was easier to use to evaluate the expression over the other.

Answers will vary. Order of operations is easier for me to visualize the steps of the expression.



For this activity, consider beginning by having students work with a partner to review the work of each student. Have them discuss what they notice and what they wonder about each method.

Both Questions 1 and 2 can be done through a class discussion forum. Instead of having students write their answers, have them discuss with their partner and facilitate a whole-class review of each one.



- Would you determine the answer using a different set of steps?
- Why do the nested grouping symbols make it possible to determine the value for $26 \div 13$ in the first step? (The division can be found because it would have to be done before the addition to the value that is produced from the nested grouping.)

11.9.3 Activity: Expression Game

Component Overview: Students will work in groups to complete a game where they roll dice to fill in the blanks an expression with the values provided by the dice. For this activity, students should work in small groups to complete.



11.9.3 Activity: Expression Game

- Each group will use a standard number cube with sides numbered 1 – 6 to play a game. One player rolls the number cube. Each player decides which box to place that number in the expression. Continue rolling the cube until all blanks in the expression are filled. Evaluate your expression and record your score based on the point system in the table. If on any round two players' expressions have the same value, both players **lose 10 points**.

Answers will vary for this activity.

Round 1 Scoring – the player with the greatest value earns 8 points.	$[(_\times_\^2)+_\]\div_\$	My answer Points earned
Round 2 Scoring – the player with the least value earns 5.5 points.	$[(_\times_\)^2+_\]\div_\$	My answer Points earned
Round 3 Scoring – the player with the greatest absolute value loses 3.7 points.	$(_\times_\^2)+(_\div_\)$	My answer Points earned
Round 4 Scoring – the player with the absolute value closest to zero earns 11 points.	$(_\times_\)^2+(_\div_\)$	My answer Points earned



If no dice are available, consider using an online random number generator, spinners, or drawing precut numbers (one through six) from a plastic bag. The key feature of this activity is for the numbers to be random so that they can get additional practice with evaluating expressions using any value.



For a challenge, consider having precut rational numbers in a plastic bag. Have students make at least one or two of their values come from the bag instead of the whole numbers from the dice.

11.9.3 Activity: Expression Game (continued)

My Game Score

Round 1	Round 2	Round 3	Round 4	My Score



As a way to close out this activity, allow select students to share out their answer to the last question. This can help settle the class back down and tie back in the mathematics prior to moving onto the next activity.

2. Discuss with the other players what strategies they used to determine where to put the numbers. Summarize your discussion.

Answers will vary based on the numbers rolled on the dice by each student.

11.9.4 Try It: Evaluating Expressions with Multiple Steps

Component Overview: Students will practice using the order of operations in order to evaluate expressions, including the steps and the explanation of each step. For this activity, students should work independently to practice what they have been learning over the past two lessons.



11.9.4 Try It: Evaluating Expressions with Multiple Steps

1. In the table an expression has been evaluated, but the steps are mixed up. Order the expressions using the numbers 1 – 7 on the left. Write an explanation for each step.

Step	Expression	Explain the Step
3	$ -11.591 + 15 - (-3.1)^2$	Evaluate inside the absolute value grouping symbol
5	$11.591 + 15 - 9.61$	Evaluate the exponent
4	$11.591 + 15 - (-3.1)^2$	Find the absolute value
1	$ 18.2 + (-3.1)^3 + 15 - (-3.1)^2$	Original expression
6	$26.591 - 9.61$	Evaluate from left to right- addition
2	$ 18.2 + (-29.791) + 15 - (-3.1)^2$	Evaluate the exponent in the absolute value grouping
7	16.981	Subtract the values for final answer



Consider taking this table and cutting it into strips with the step numbers, the expressions, and a premade explanation. Allow students to work in partner pairs to sort them out and try to match them in the correct order for the table.

The expression $|18.2 + (-3.1)^3| + 15 - (-3.1)^2$ can be evaluated differently than using the steps shown. For example, students who have a grasp of the order of operations, properties of equality, and laws of exponents may prefer to use these steps:

$$\begin{aligned}
 &|18.2 + -29.791| + 15 - 9.61 \\
 &|-11.591| + 5.39 \\
 &11.591 + 5.39 \\
 &16.981
 \end{aligned}$$

The intent is not for students to be forced to do one step at a time when evaluating expressions but to ensure they can explain the steps. Alternatively, your students could show their own work and explain the mathematics that were applied.

11.9.4 Try It: Evaluating Expressions with Multiple Steps (continued)

2. Evaluate each expression without using a calculator.

a. $\left(\frac{3}{4}\right)^2 \times 32 + 3^2$
 $\frac{3^2}{4^2} \times 32 + 3^2$
 $\frac{9}{16} \times 32 + 9$
 $18 + 9$
 27

b. $[(10 + 4.5)^2 - 10.25]^2 \div 40$
 $[(14.5)^2 - 10.25]^2 \div 40$
 $[210.25 - 10.25]^2 \div 40$
 $200^2 \div 40$
 $40000 \div 40$
 1000

c. $6.7 - (2^3 - 3)^3 \div 10 + 4$
 $6.7 - (8 - 3)^3 \div 10 + 4$
 $6.7 - 5^3 \div 10 + 4$
 $6.7 - 125 \div 10 + 4$
 $6.7 - 12.5 + 4$
 $-5.8 + 4$
 -1.8



When monitoring student performance on question 2, watch for their steps more than the final answer. It is probable that students may make a calculation error without their calculator, but it is important that their process be correct.

11.9.5 Activity: Class Debate

Component Overview: Students will engage in a class debate over choosing an expression that will result in a specific value. This activity is for the entire class but initially done independently.



11.9.5 Activity: Class Debate

James wants to choose an expression that has a value of -778 . He can choose between the expressions shown.

Expression A	Expression B
$14 - (10 \times 3^2)^2 + 6 \times 3$	$14 - 10 \times (3^2)^2 + 6 \times 3$
Expression C	Expression D
$(14 - 10) \times (3^2 + 6)^2 \times 3$	$14 - [10 \times (3^2)^2] + 6 \times 3$

Which expression should James choose?

Expressions B and D are equivalent to -778 .

- Each corner of the room is labeled A, B, C, or D. After selecting the expression that you think James should choose, go to the corner of the room that matches the label for that expression.
- With your classmates who selected the same expression, discuss why your expression is the expression that James should choose. Discuss why James should not choose the other expressions. One member of your group will act as a representative during a class debate.
Answers will vary.
- For the debate, prepare an opening statement and potential rebuttals for arguments from other groups. Keep arguments and rebuttals centered on the mathematics.
Answers will vary.
- After the debate, explain whether you would select the same expression again.
Answers will vary.



Prior to beginning this activity, consider reviewing the instructions with your students before releasing them to complete.

Students should be instructed to try and determine their own answer within a given time frame. Then, once time is called, they can complete the remaining parts of the activity that are listed in the boxes.



Consider adding an additional layer of engagement. Allow the group members to move to a different corner (if they think they should) after each presentation to see if anyone has convinced others to join their expression.

11.9.6 Wrap-Up: Ticket Out the Door

Component Overview: Students will complete a ticket out the door on their learning from this lesson. This activity should be done independently to understand each student's level of proficiency.



11.9.6 Wrap-Up: Ticket Out the Door

- Evaluate each expression without a calculator. Show each step you use.

$$-\left|5 - \frac{3}{5}\right| + \frac{4}{15}(4^2 - 61)$$

Work

$$\begin{aligned} &= -\left|\frac{22}{5}\right| + \frac{4}{15}(4^2 - 61) \\ &= -\frac{22}{5} + \frac{4}{15}(4^2 - 61) \\ &= -\frac{22}{5} + \frac{4}{15}(16 - 61) \\ &= -\frac{22}{5} + \frac{4}{15}(-45) \\ &= -\frac{22}{5} - 12 \\ &= -\frac{82}{5} \end{aligned}$$

$$\frac{(1.5 + |-3.25 - 4.25|)^3}{12}$$

Work

$$\begin{aligned} &= \frac{(1.5 + |-7.5|)^3}{12} \\ &= \frac{(1.5 + 7.5)^3}{12} \\ &= \frac{(9)^3}{12} \\ &= \frac{729}{12} \\ &= \frac{729}{12} \text{ or } \frac{243}{4} \end{aligned}$$



Proficiency on this Wrap-Up includes:

- Understanding which operations should be done first. Students may choose to perform more than one operation first by considering the properties of equality.
- Understanding that one value should be determined from an expression inside an absolute value or a set of parentheses.
- Fluency of operations with rational numbers. A small miscalculation that a student carries on throughout their work should be considered a minor mathematical error. Fluency can be shown despite a small miscalculation. Teachers are encouraged to consider a student's work when determining fluency.